

- 1 The binomial expansion of $\left(2x + \frac{5}{x}\right)^6$ has a term which is a constant. Find this term. [4]
- 2 Find the coefficient of x^3 in the binomial expansion of $(2 - 4x)^5$. [4]
- 3 Find the coefficient of x^4 in the binomial expansion of $(5 + 2x)^7$. [4]
- 4
- (i) An arithmetic progression has first term A and common difference D . The sum of its first two terms is 25 and the sum of its first four terms is 250.
- A Find the values of A and D . [4]
- B Find the sum of the 21st to 50th terms inclusive of this sequence. [3]
- (ii) A geometric progression has first term a and common ratio r , with $r \neq \pm 1$. The sum of its first two terms is 25 and the sum of its first four terms is 250.
- Use the formula for the sum of a geometric progression to show that $\frac{r^4 - 1}{r^2 - 1} = 10$ and hence or otherwise find algebraically the possible values of r and the corresponding values of a . [5]
- 5 Find the binomial expansion of $(1 - 5x)^4$, expressing the terms as simply as possible. [4]

- 6 For each of the following sequences, state with a reason whether it is convergent, periodic or neither. Each sequence continues in the pattern established by the given terms.

(i) $3, \frac{3}{2}, \frac{3}{4}, \frac{3}{8}, \dots$

[1]

(ii) $3, 7, 11, 15, \dots$

[1]

(iii) $3, 5, -3, -5, 3, 5, -3, -5, \dots$

[1]

- 7 S is the sum to infinity of a geometric progression with first term a and common ratio r .

(i) Another geometric progression has first term $2a$ and common ratio r . Express the sum to infinity of this progression in terms of S .

[1]

(ii) A third geometric progression has first term a and common ratio r^2 . Express, in its simplest form, the sum to infinity of this progression in terms of S and r .

[2]

- 8 The n th term of a sequence, u_n , is given by

$$u_n = 12 - \frac{1}{2}n.$$

(i) Write down the values of u_1 , u_2 and u_3 . State what type of sequence this is.

[2]

(ii) Find $\sum_{n=1}^{30} u_n$.

[3]

- 9 The second term of a geometric progression is 24. The sum to infinity of this progression is 150. Write down two equations in a and r , where a is the first term and r is the common ratio. Solve your equations to find the possible values of a and r .

[5]

(i) Find $\sum_{r=1}^5 \frac{21}{r+2}$

[2]

(ii) A sequence is defined by

$$u_1 = a, \text{ where } a \text{ is an unknown constant,}$$
$$u_{n+1} = u_n + 5.$$

Find, in terms of a , the tenth term and the sum of the first ten terms of this sequence.

[3]



- 11 Jill has 3 daughters and no sons. They are generation 1 of Jill's descendants.

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Each of her daughters has 3 daughters and no sons. Jill's 9 granddaughters are generation 2 of her descendants. Each of her granddaughters has 3 daughters and no sons; they are descendant generation 3.

Jill decides to investigate what would happen if this pattern continues, with each descendant having 3 daughters and no sons.

- (i) How many of Jill's descendants would there be in generation 8?

[2]

- (ii) How many of Jill's descendants would there be altogether in the first 15 generations?

[3]

- (iii) After n generations, Jill would have over a million descendants altogether. Show that n satisfies the inequality

$$n > \frac{\log_{10} 2000003}{\log_{10} 3} - 1.$$

Hence find the least possible value of n .

[4]

- (iv) How many **fewer** descendants would Jill have altogether in 15 generations if instead of having 3 daughters, she and each subsequent descendant has 2 daughters?

[3]

- 12 An arithmetic progression has tenth term 11.1 and fiftieth term 7.1. Find the first term and the common difference. Find also the sum of the first fifty terms of the progression.

[5]

13

A sequence is defined by $u_1 = 2$ and $u_{k+1} = \frac{10}{u_k^2}$.

Calculate $\sum_{k=1}^4 u_k$.

[3]

- 14 An arithmetic progression (AP) and a geometric progression (GP) have the same first and fourth terms as each other. The first term of both is 1.5 and the fourth term of both is 12. Calculate the difference between the tenth terms of the AP and the GP.

[5]

- 15 A sequence is defined as follows.

$$u_1 = a, \text{ where } a > 0$$

To obtain u_{r+1}

- find the remainder when u_r is divided by 3,
- multiply the remainder by 5,
- the result is u_{r+1} .

Find $\sum_{r=2}^4 u_r$ in each of the following cases.

- (i) $a = 5$
(ii) $a = 6$

[3]

- 16 Find the first four terms of the binomial expansion of $\sqrt[3]{1-2x}$. State the set of values of x for which the expansion is valid.

[6]

17

- (i) Express $\frac{x}{(1+x)(1-2x)}$ in partial fractions.

[3]

- (ii) Hence use binomial expansions to show that $\frac{x}{(1+x)(1-2x)} = ax + bx^2 + \dots$, where a and b are constants to be determined.

State the set of values of x for which the expansion is valid.

[5]

- 18 Find the first three terms in the binomial expansion of $(4+x)^{\frac{3}{2}}$. State the set of values of x for which the expansion is valid.

[5]

19

- (i) Find the first three terms of the binomial expansion of $\frac{1}{\sqrt[3]{1-2x}}$. State the set of values of x for which the expansion is valid.

[5]

- (ii) Hence find a and b such that $\frac{1-3x}{\sqrt[3]{1-2x}} = 1 + ax + bx^2 + \dots$.

[3]

- 20 Given that $\left(1 + \frac{x}{p}\right)^q = 1 - x + \frac{3}{4}x^2 + \dots$, find p and q , and state the set of values of x for which the expansion is valid.

[7]

- 21 Find the coefficient of x^4 in the binomial expansion of $(x-3)^5$.

[3]

- 22 A geometric series has first term 3. The sum to infinity of the series is 8.
Find the common ratio.

[3]

23

Find the first four terms of the binomial expansion of $(1 - 2x)^{\frac{1}{2}}$.

[4]

State the set of values of x for which the expansion is valid.

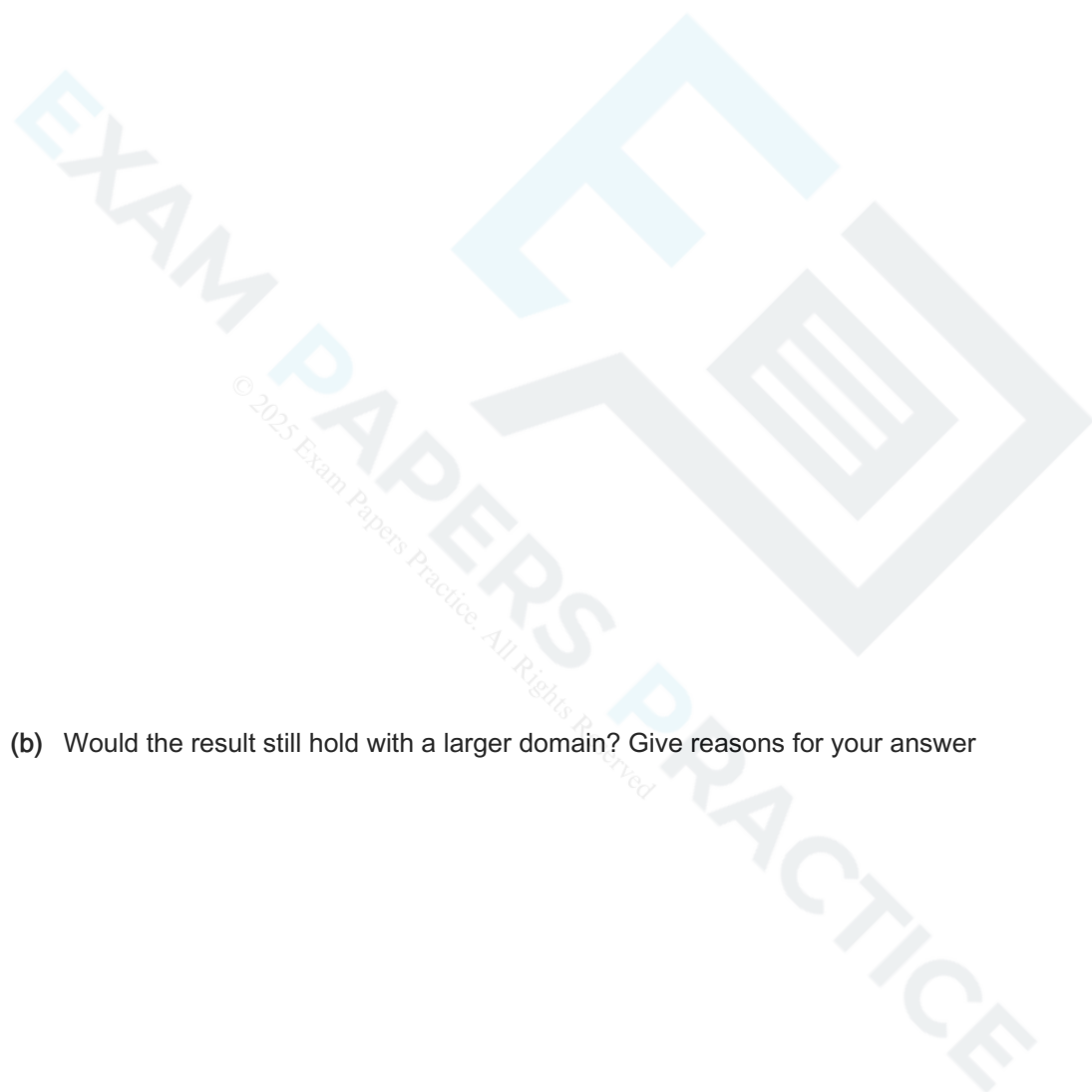
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Show that $\sum_{r=1}^4 \ln \frac{r}{r+1} = -\ln 5$.

25 The curve $y = f(x)$ is defined by the function $f(x) = e^{-x} \sin x$ with domain $0 \leq x \leq 4\pi$.

- (a) (i) Show that the x -coordinates of the stationary points of the curve $y = f(x)$, when arranged in increasing order, form an arithmetic sequence.

- (ii) Show that the corresponding y -coordinates form a geometric sequence.



(b) Would the result still hold with a larger domain? Give reasons for your answer

[1]

26 Expand $(2x - 3)^5$, writing each term in its simplest form.

[4]

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27 The sequence of positive numbers $a_1, a_2, a_3, \dots$ is a geometric sequence. Prove that the sequence $\ln a_1, \ln a_2, \ln a_3, \dots$ is an arithmetic sequence.

[3]

28 (a) Use the binomial expansion to show that $(1 - 2x)^{-\frac{1}{2}} \approx 1 + x + \frac{3}{2}x^2$ for sufficiently small values of x .

[2]

(b) For what values of x is the expansion valid?

[1]

(c) Find the expansion of $\sqrt{\frac{1+2x}{1-2x}}$ in ascending powers of x as far as the term in x^2 .

[3]

(d) Use $x = \frac{1}{20}$ in your answer to part (c) to find an approximate value for $\sqrt{11}$.

[2]

29 Prove from first principles that the derivative of x^3 is $3x^2$.

[5]

30 In this question you must show detailed reasoning.

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A geometric series has first term $(b^2 - 13)$, common ratio $\frac{1}{b}$ and sum to infinity -6 .

Find the possible values of the common ratio.

[9]



- 31 Tom is carrying out a survey into the way in which students travel to school. He selects 50 students and asks each of them 'How did you get to school this morning?' The results are given in the table below.

Walk	Cycle	Bus	Car
17	9	13	11

Tom then randomly selects 4 of these students to interview in more detail.

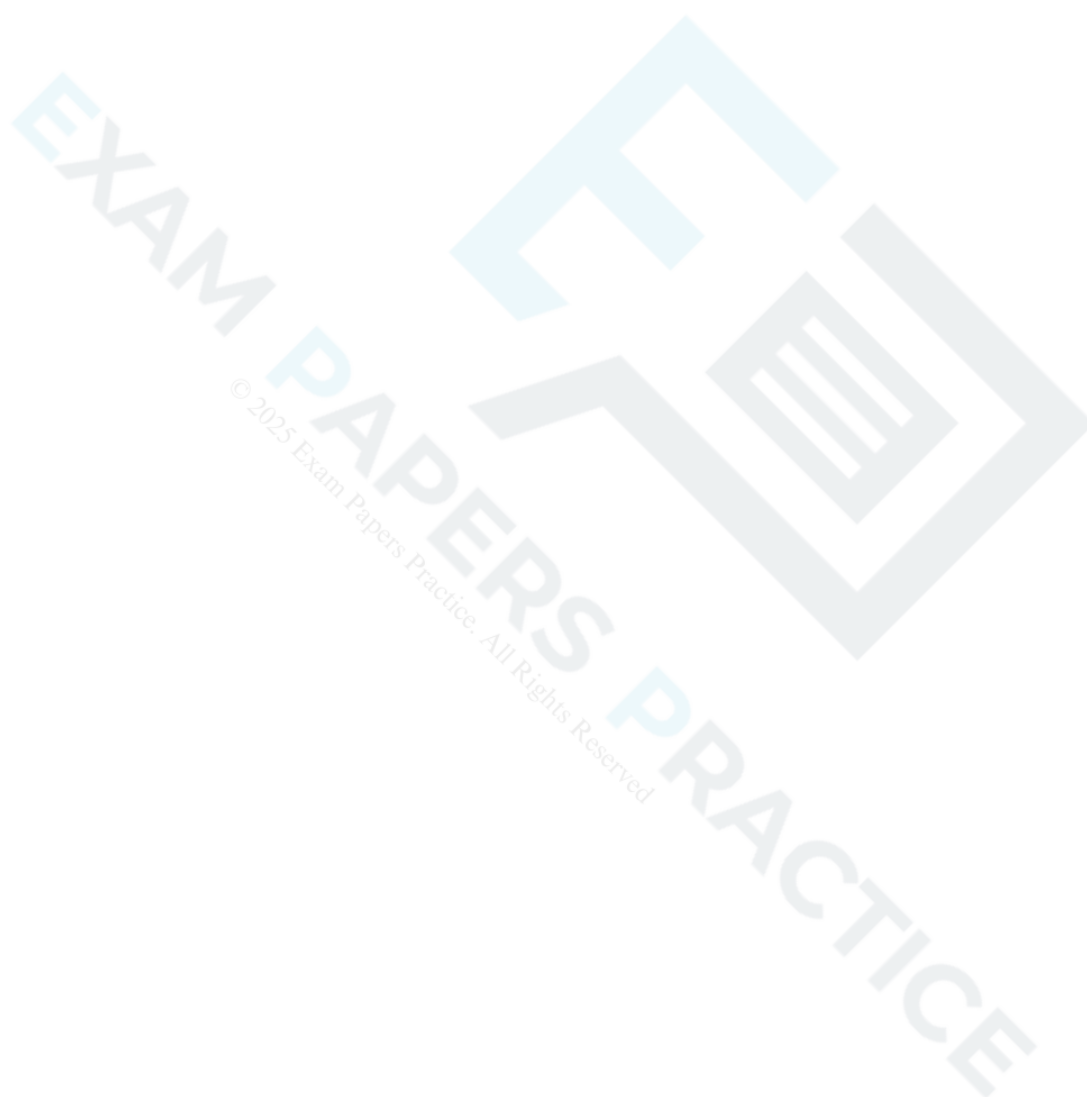
- (i) Find the number of ways in which Tom can select the 4 students. [2]

- (ii) Find the probability that all 4 of these students walked to school. [2]

- (iii) Find the probability that at least 2 of the 4 students used the same method to get to school. [4]

- 32 You are given that, in the expansion of $(a + bx)^5$, the constant term is 32 and the coefficient of x^3 is -1080 . Find the values of a and b .

[5]



- (i) Calculate $\sum_{r=1}^5 (3r+2)$.

[2]

- (ii) An arithmetic progression (AP) has first term 4.2 and sixth term 1.8. Find the common difference of this AP.

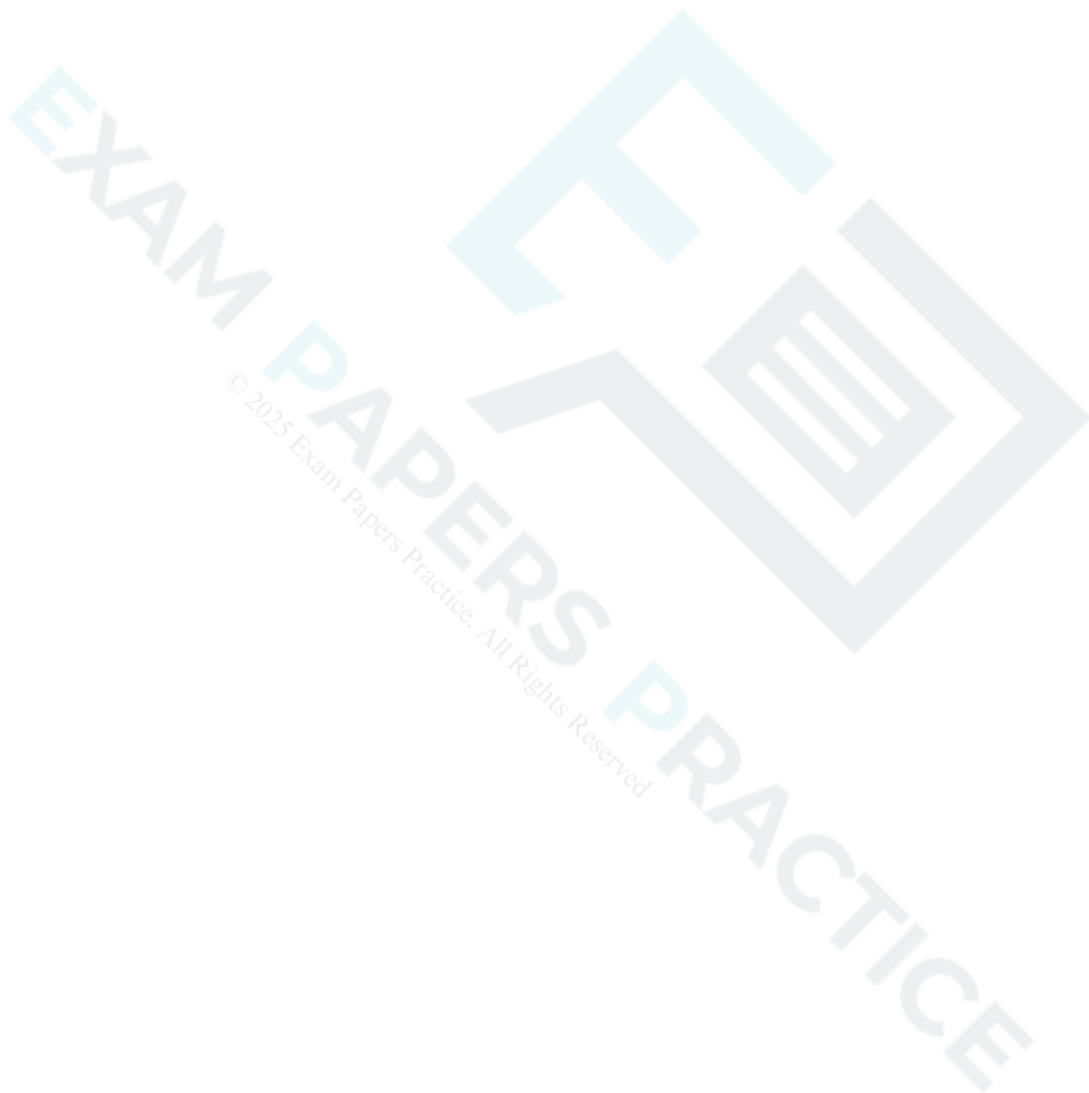
[2]

34 A firm takes on two new employees, Arif and Bettina.

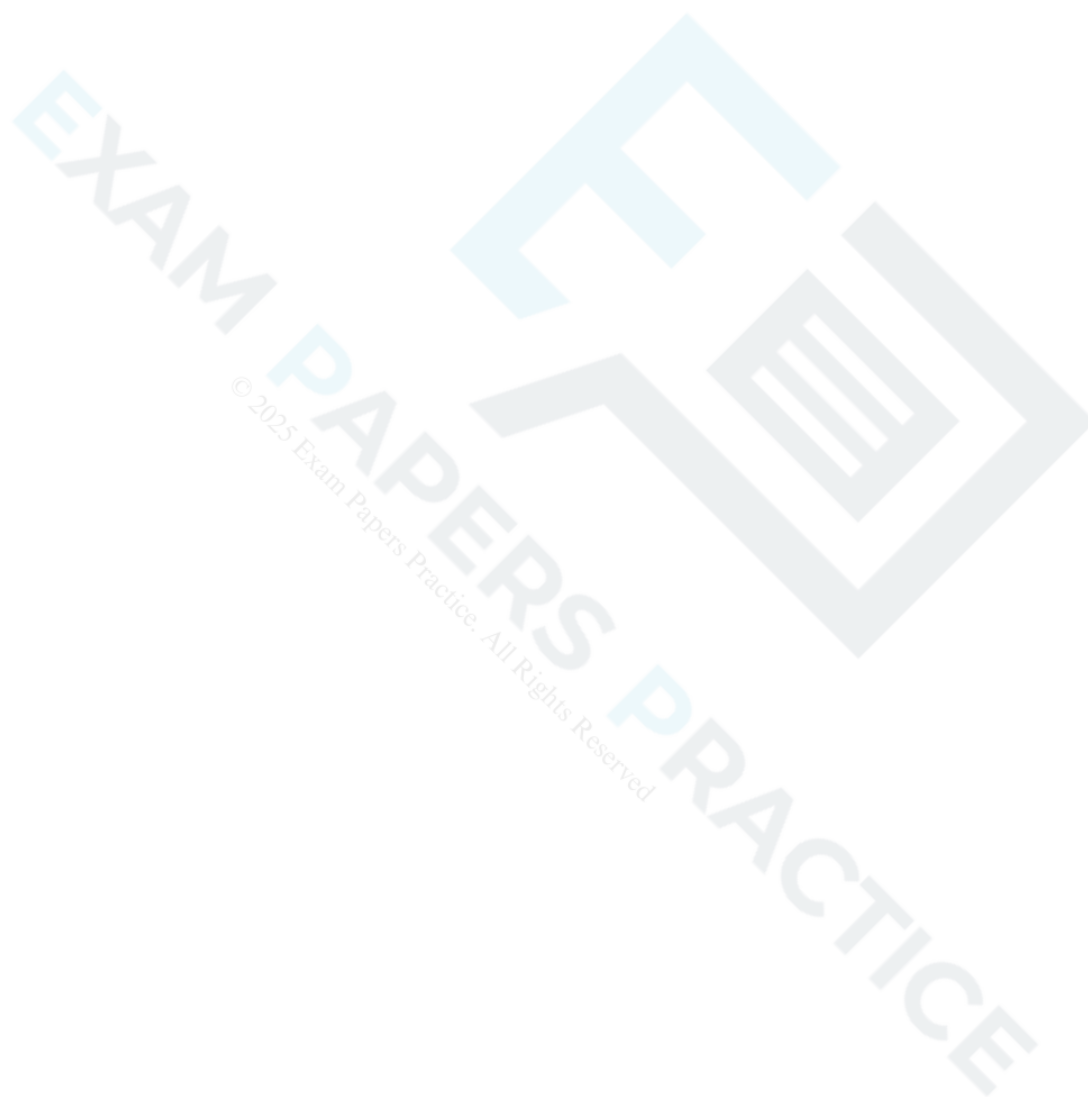
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- Arif starts on an annual salary of £30 000, and his salary increases by £1000 each year after that.
 - Bettina starts on an annual salary of £25 000, and her salary then increases by 5% each year after that. (So, for example, Bettina's salary in year 3 is 5% greater than her salary in year 2.)
- (i) Show that Arif earns more than Bettina in year 10 of their employment, but Arif earns less than Bettina in year 11.

[4]



- (ii) Show that the total amounts earned by each of Arif and Bettina during their employment up to the end of year 17, correct to the nearest £100, are equal. [4]

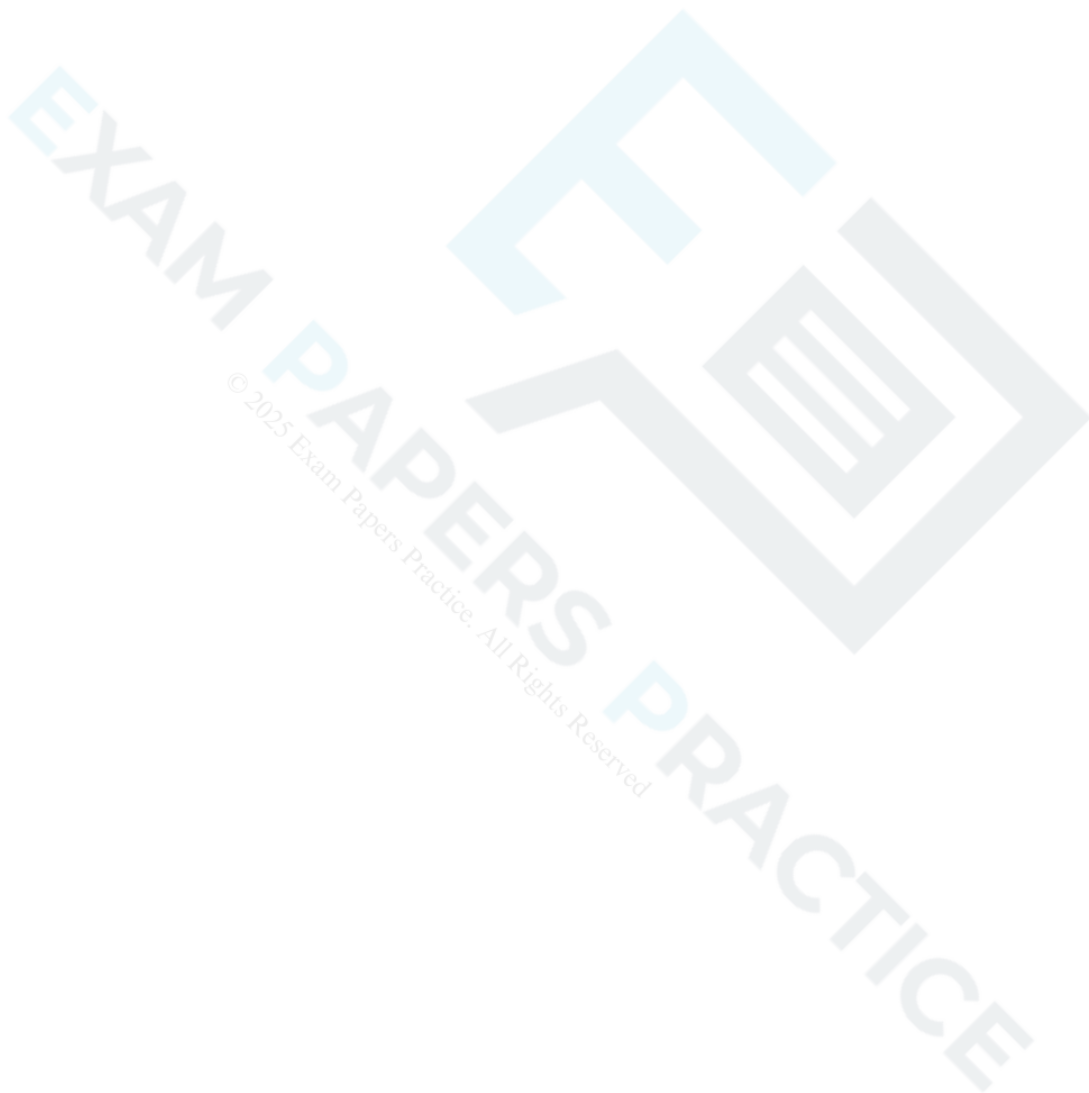


- (iii) At the end of year n , the total that Bettina has earned during this employment is greater than £ M .

Show that $n > \frac{\log_{10}(M + 500\,000) - \log_{10} 500\,000}{\log_{10} 1.05}$

Hence find in which year the total that Bettina has earned during this employment is first greater than £1.2 million.

[5]





- (i) Express $\frac{5-x}{(2-x)(1+x)}$ in partial fractions.

[3]

- (ii) Hence or otherwise find the first 3 terms of the binomial expansion of $\frac{5-x}{(2-x)(1+x)}$ in ascending powers of x .

[5]

36 In this question you must show detailed reasoning.

Given that

$$(1 + ax)^n = 1 + 6x - 6x^2 + \dots,$$

where a and n are constants, find the values of a and n .

[6]



37 (a) Find the sum of all the even numbers from 2 to 1000.

[3]

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(b) The sum of n consecutive even numbers, starting at 2, is less than 110.

(i) Show that $n^2 + n < 110$.

[1]

(ii) Find the set of possible values of n given that $n \neq 0$.

[3]

(a) Show how the substitution $x = y - \frac{m}{3y}$ can be used to transform $x^3 + mx = n$ into a quadratic equation in y^3 . [3]

(b) Show that, when $m > 0$, the resulting quadratic equation in y^3 has distinct real roots. [2]

- 39 Aleela and Baraka are saving to buy a car. Aleela saves £50 in the first month. She increases the amount she saves by £20 each month.

(a) Calculate how much she saves in two years.

[2]

Baraka also saves £50 in the first month. The amount he saves each month is 12% more than the amount he saved in the previous month.

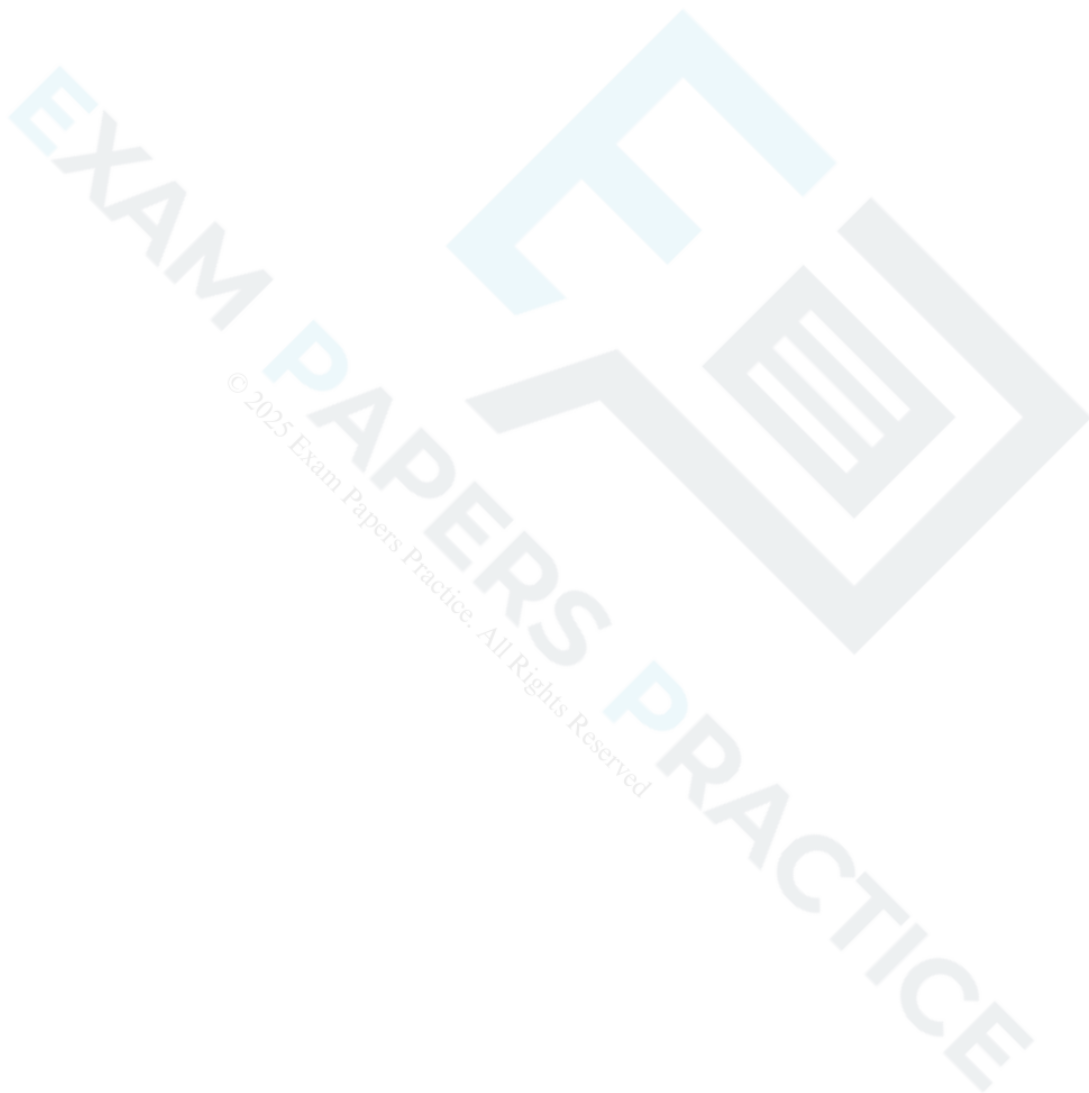
(b) Explain why the amounts Baraka saves each month form a geometric sequence.

[1]



(c) Determine whether Baraka saves more in two years than Aleela.

[3]





The function $f(x)$ is defined by $f(x) = \sqrt[3]{27-8x^3}$. Jenny uses her scientific calculator to create a table of values for $f(x)$ and $f'(x)$.

x	$f(x)$	$f'(x)$
0	3	0
0.25	2.9954	-0.056
0.5	2.9625	-0.228
0.75	2.8694	-0.547
1	2.6684	-1.124
1.25	2.2490	-1.977
1.5	0	ERROR

- (a) Use calculus to find an expression for $f'(x)$ and hence explain why the calculator gives an error for $f'(1.5)$.

[3]

- (b) Find the first three terms of the binomial expansion of $f(x)$.

[3]

- (c) Jenny integrates the first three terms of the binomial expansion of $f(x)$ to estimate the value of $\int_0^1 \sqrt[3]{27-8x^3} \, dx$. Explain why Jenny's method is valid in this case. (You do not need to evaluate Jenny's approximation.)

[2]

- (d) Use the trapezium rule with 4 strips to obtain an estimate for $\int_0^1 \sqrt[3]{27-8x^3} \, dx$.

[3]

The calculator gives 2.921 174 38 for $\int_0^1 \sqrt[3]{27-8x^3} dx$. The graph of $y = f(x)$ is shown in Fig. 13.

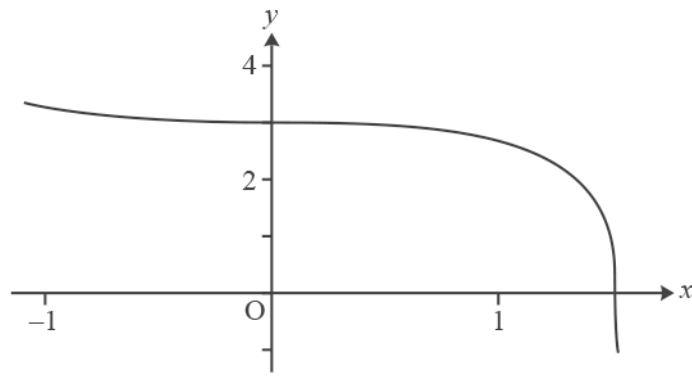


Fig. 13

(e) Explain why the trapezium rule gives an underestimate.

[1]



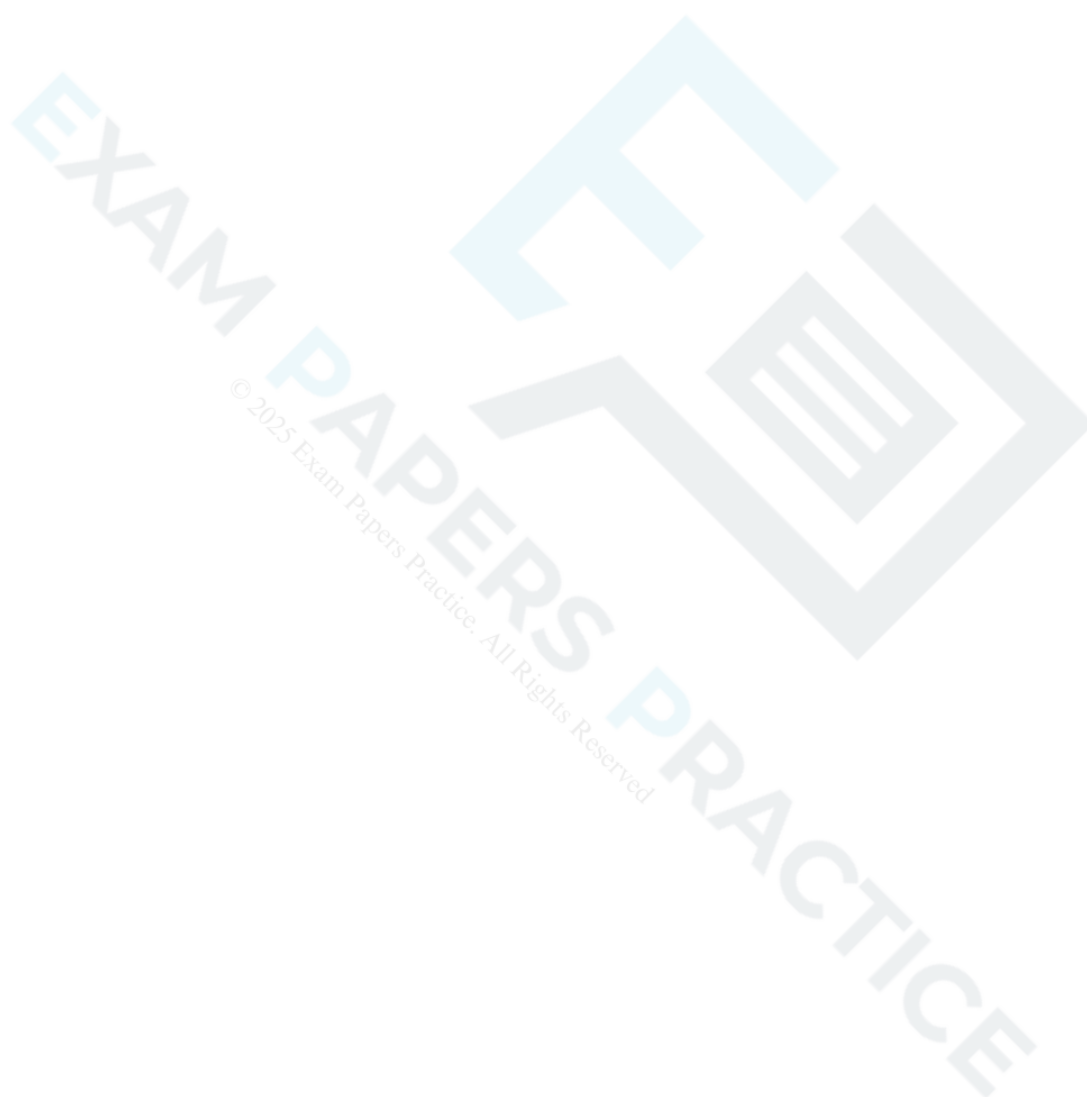
Find the constant term in the expansion of $\left(x^2 + \frac{1}{x}\right)^{15}$.

[2]



- 42 (See Insert for Jun18 64003.) Consider a geometric sequence in which all the terms are positive real numbers. Show that, for any three consecutive terms of this sequence, the middle one is the geometric mean of the other two.

[3]



43 Find the binomial expansion of $(3 - 2x)^3$.

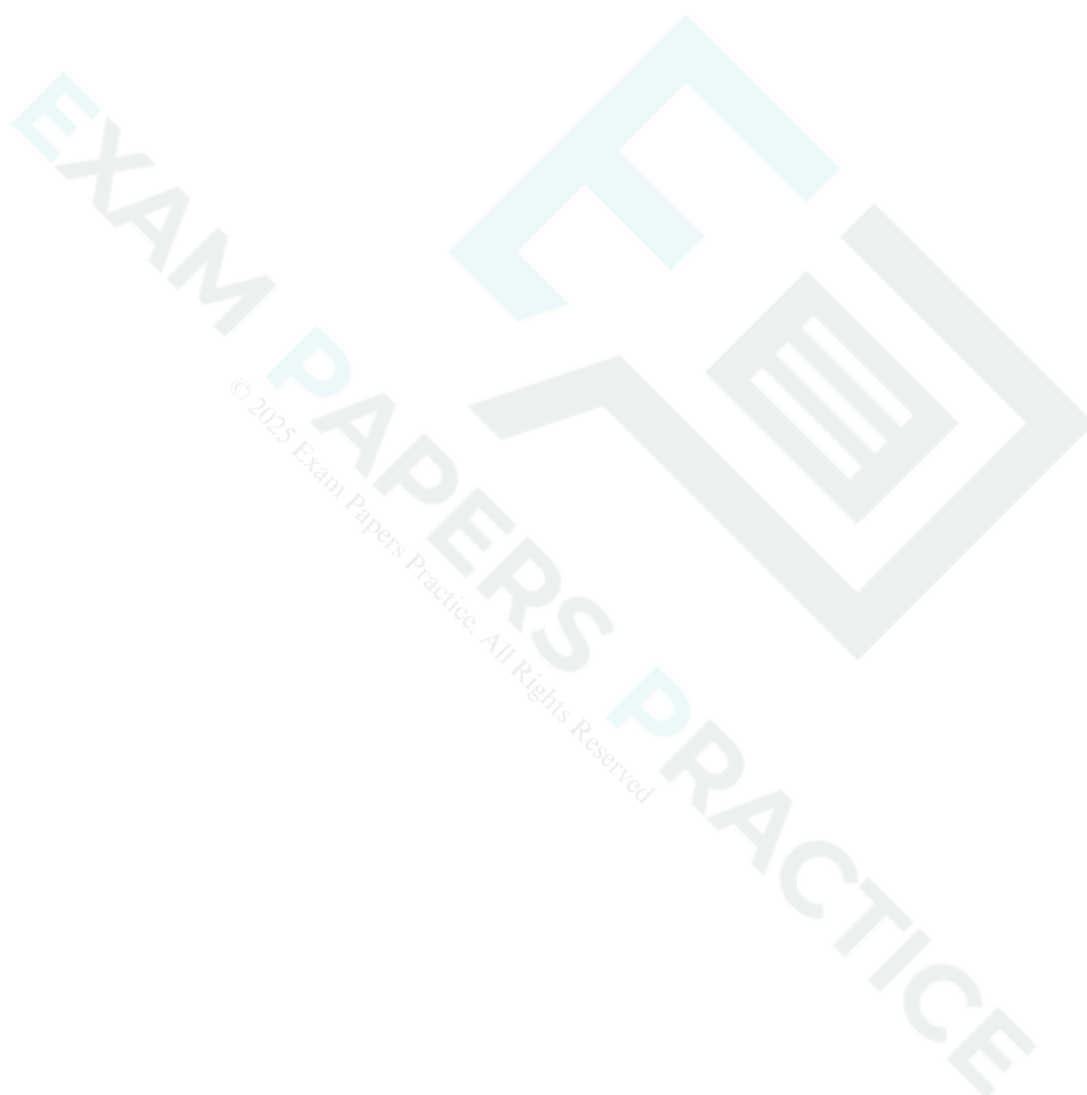
[4]

44 Find the term in x^3 in the binomial expansion of $(2 + x)^5$.

[2]

45 An arithmetic sequence has third term 6 and ninth term 30. Find the sum of the first 100 terms.

[4]



- 46 State, with a reason, whether each of the following sequences are convergent, divergent, periodic, or none of these.

(a) $u_n = \sin\left(\frac{n\pi}{2}\right), n = 1, 2, 3, \dots$ [1]

(b) 100, 150, 175, 187.5, 193.75, 196.875, ... [1]

(c) $u_{k+1} = (-2)^k u_k, u_1 = 3$ [1]

- (a) Find the first four terms in the expansion of $\left(1 + \frac{x}{2}\right)^{-2}$.

[3]

- (b) State the range of values of x for which this expansion is valid.

[1]

48 The 5th term of an arithmetic progression is 11 and the 11th term of the progression is 20. Find

- the common difference,
- the first term,
- the sum of the first 40 terms of the progression.

[5]



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49 (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $(1 + 3x)^{-1}$

[3]

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(b) State the range of values of x for which this expansion is valid.

[1]

50 (See Insert for H640/03, Practice 4.)

Assume that the length of each side of the equilateral triangle shown in Fig. C1.1 is one unit.

(a) Find the perimeter of the second iteration, shown in Fig. C1.3.

[1]

(b) Find an expression for the perimeter of the n th iteration.

[3]

(c) Show that, as stated in line 11, the perimeter of the Koch snowflake is not finite.

[1]

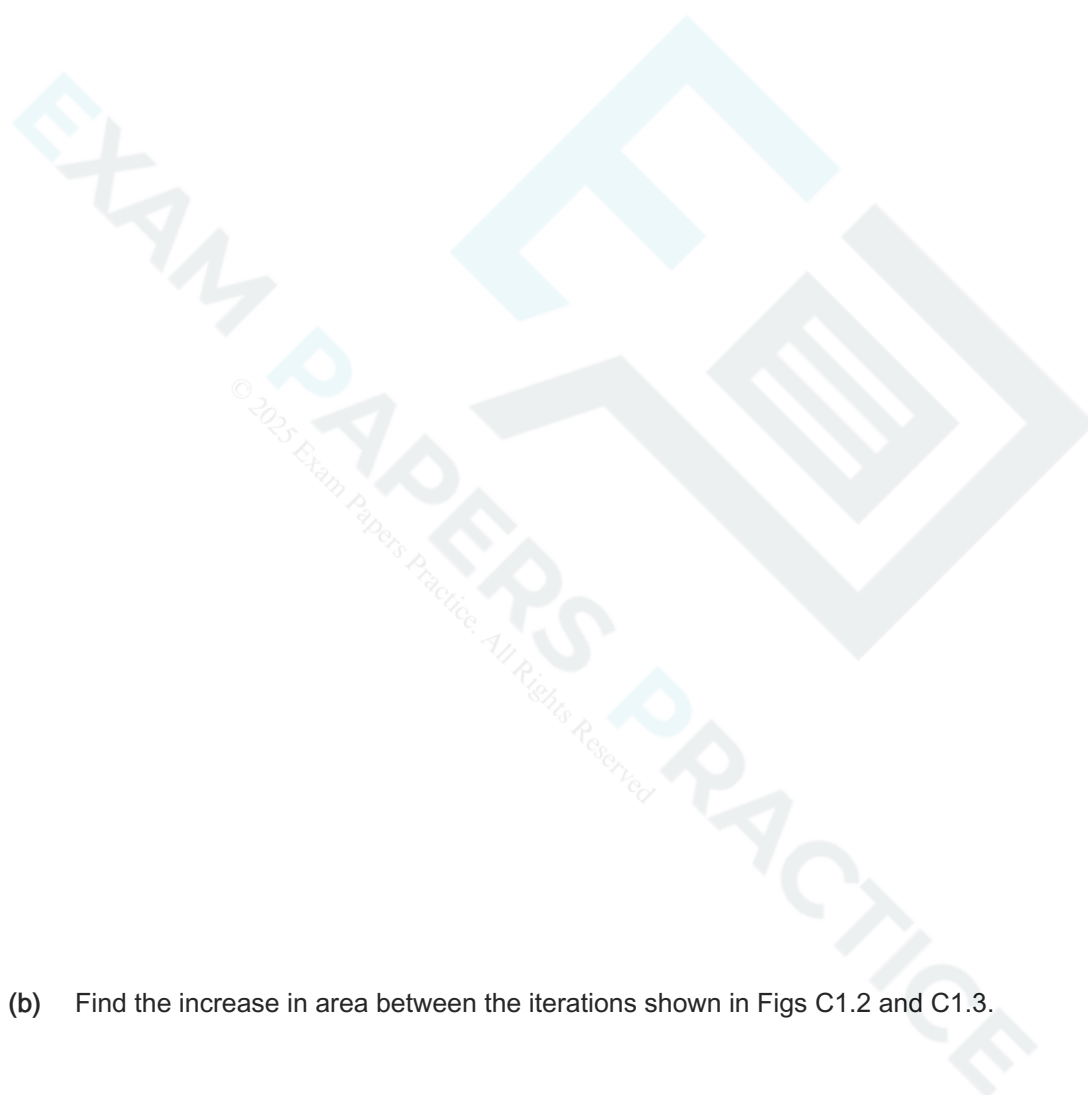
51 (See Insert for H640/03, Practice 4.)



Assume now that the area of the equilateral triangle shown in Fig. C1.1 is one unit of area.

(a) Find the area of the first iteration, shown in Fig. C1.2.

[2]



(b) Find the increase in area between the iterations shown in Figs C1.2 and C1.3.

[1]

(c) Find the area of the Koch snowflake.

[4]

52(a) The function $f(x)$ is given by $f(x) = (1 - \alpha x)^{-3}$, where α is a non-zero constant. In the binomial expansion of $f(x)$, the coefficients of x and x^2 are equal.

Find the value of α .

[3]

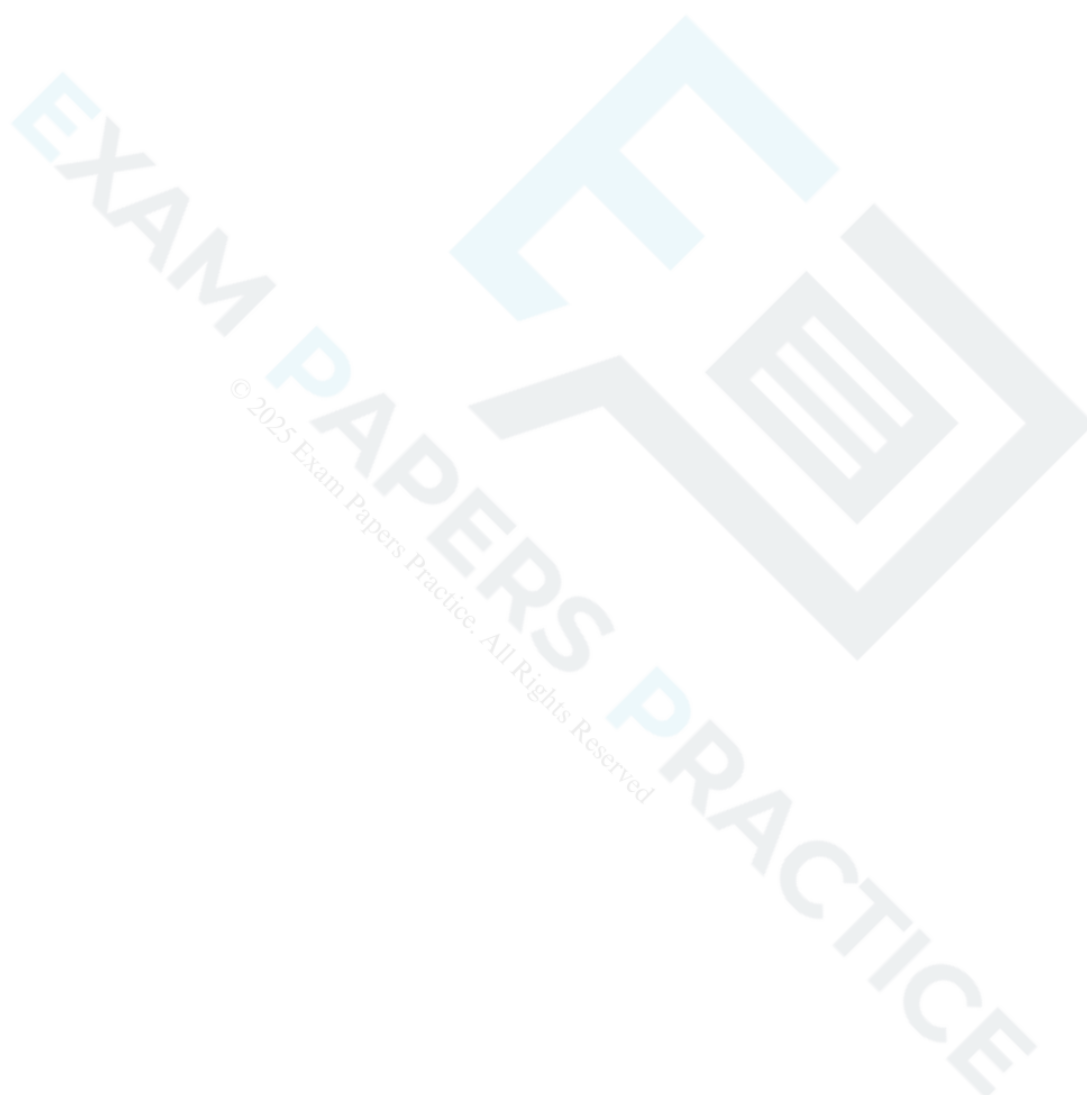
(b) Using this value for α ,

(i) state the set of values of x for which the binomial expansion is valid,

[1]

(ii) write down the quadratic function which approximates $f(x)$ when x is small.

[1]



53(a) The population of Melchester is 185 207. During a nationwide flu epidemic the number of new cases in Melchester are recorded each day. The results from the first three days are shown in Fig. 13.

Day	1	2	3
Number of new cases	8	24	72

Fig. 13

A doctor notices that the numbers of new cases on successive days are in geometric progression. Find the common ratio for this geometric progression.

[1]

- (b) The doctor uses this geometric progression to model the number of new cases of flu in Melchester.

According to the model, how many new cases will there be on day 5?

[1]

- (c) Find a formula for the **total** number of cases from day 1 to day n inclusive according to this model, simplifying your answer.

[1]

- (d) Determine the maximum number of days for which the model could be viable in Melchester.

[3]

- (e) State, with a reason, whether it is likely that the model will be viable for the number of days found in the above part.

[1]



By using the substitution $u = 1 + \sqrt{x}$, find $\int \frac{x}{1 + \sqrt{x}} dx$.

[6]



55(a) An arithmetic series has first term 9300 and 10th term 3900.

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Show that the 20th term of the series is negative.

[3]

(b) The sum of the first n terms is denoted by S . Find the greatest value of S as n varies.

[4]

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56 A function is defined by $f(x) = x^3 - x$.

By considering $\frac{f(x+h) - f(x)}{h}$, show from first principles that $f'(x) = 3x^2 - 1$.

[4]

57

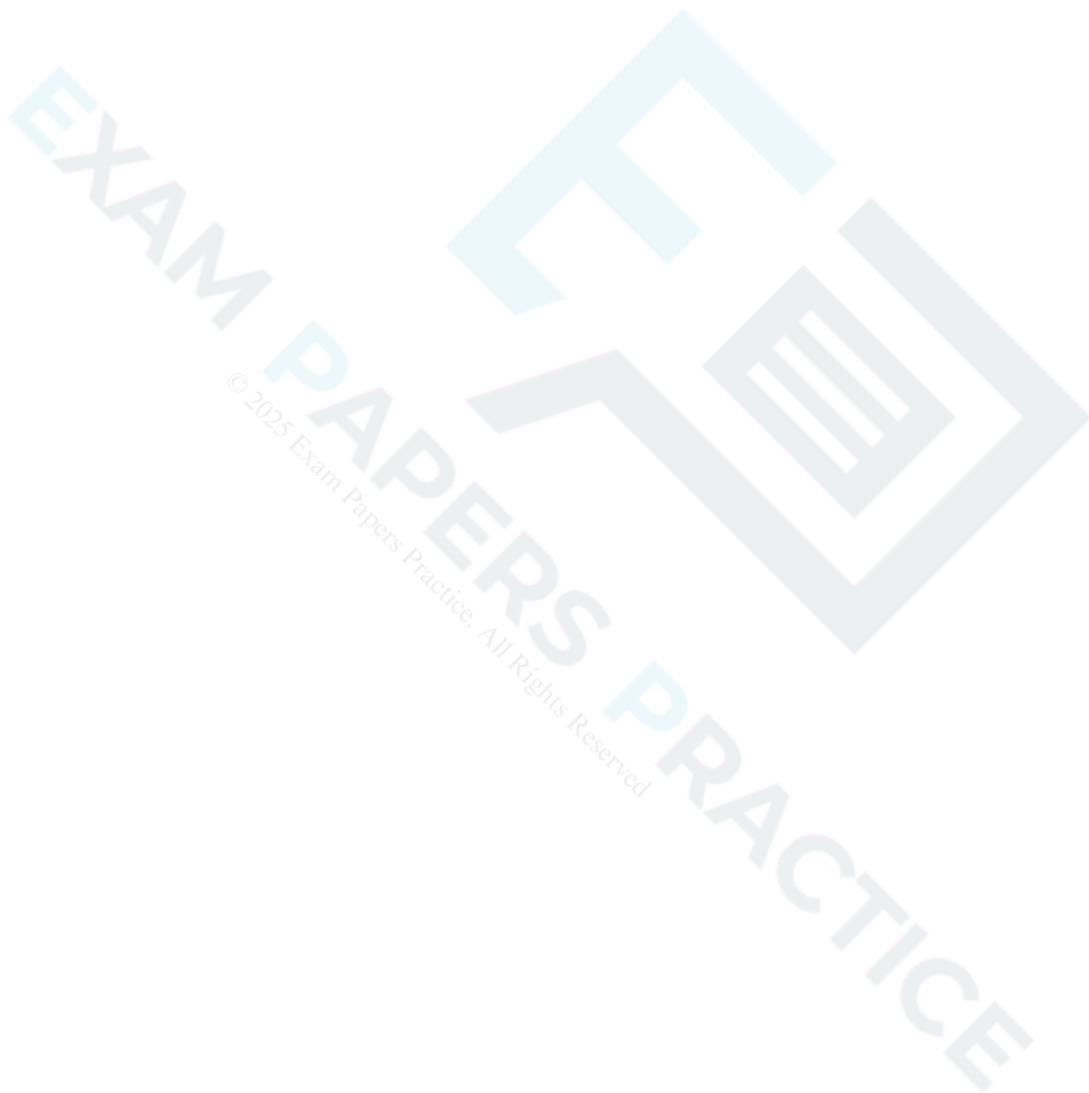
Find the value of $\sum_{r=1}^5 2^r(r-1)$.

[2]

- 58 A particular phone battery will last 10 hours when it is first used. Every time it is recharged, it will only last 98% of its previous time.

Find the maximum total length of use for the battery.

[3]



59(a) The first n terms of an arithmetic series are

$$17 + 28 + 39 + \dots + 281 + 292.$$

Find the value of n .

[1]

(b) Find the sum of these n terms.

[2]

60(a)

Find the first three terms in ascending powers of x of the binomial expansion of $(1 + 4x)^{\frac{1}{2}}$.

[3]

(b) State the range of values of x for which this expansion is valid.

[1]

END OF QUESTION PAPER

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Question			Answer/Indicative content	Marks	Part marks and guidance
			20000	A1	or B4 for 20 000 obtained from multiplying out $\left(2x + \frac{5}{x}\right)^6$ allow SC3 for 20000 as part of an expansion Examiner's Comments A large proportion of candidates did not understand what was meant by 'a term which is constant'. A good number still found the term $20(2x)^3\left(\frac{5}{x}\right)^3$ but did not recognise it as the term needed to find the constant. Even those who did know what was meant by a constant term usually wrote out the whole expansion rather than identifying which was the relevant term from the start. Brackets were often missing, leading to incorrect evaluations.
			Total	4	

Question			Answer/Indicative content	Marks	Part marks and guidance	
2			-2560 www	4	B3 for 2560 from correct term (NB coefficient of x^4 is 2560) or B3 for neg answer following $10 \times 4 \times -64$ and then an error in multiplication or M2 for $10 \times 2^2 \times (-4)^3$ oe; must have multn signs or be followed by a clear attempt at multn; or M1 for $2^2 \times (-4)^3$ oe (condone missing brackets) or for 10 used or for 1 5 10 10 5 1 seen	ignore terms for other powers; condone x^3 included; but eg $10 \times 4 \times -64 = 40 - 64 = -24$ gets M2 only condone missing brackets eg allow M2 for $10 \times 2^2 \times -4x^3$ 5C_3 or factorial notation is not sufficient but accept $\frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1}$ oe 10 may be unsimplified, as above M1 only for eg 10, 2^2 and $-4x^3$ seen in table with no multn signs or evidence of attempt at multn

Question			Answer/Indicative content	Marks	Part marks and guidance	
					for those who find the coefft of x^2 instead: allow M1 for 10 used or for 1 5 10 10 5 1 seen; and a further SC1 if they get 1280, similarly for finding coefficient of x^4 as 2560 Examiner's Comments Finding the binomial coefficient was done successfully by many candidates, but a surprising number omitted the negative sign in their answer. Virtually all the candidates managed to pick up at least one mark, usually for writing down the binomial coefficient either in Pascal's triangle or as part of an expression. Many candidates wrote down an expression involving the key elements 10, 2^2 and $(-4)^3$, though the brackets were often omitted. It was at this point that some arithmetical errors crept in, in the attempts to calculate $10 \times 4 \times -64$.	[lack of neg sign in the x^2 or x^4 terms means that these are easier and so not eligible for just a 1 mark MR penalty]
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3			70 000 www	4	throughout, condone xs included eg $(2x)^4$ M3 for $35 \times 5^3 \times 2^4$ oe or M2 for two of these elements multiplied or M1 for 35 oe or for 1 7 21 35 35 21 7 1 row of Pascal's triangle seen Examiner's Comments Many candidates were able to establish the desired product of $35 \times 5^3 \times 2^2$ in finding the binomial coefficient. There were fewer failing to cope correctly with $(2x)^4$ than in similar past questions on this topic. However, few candidates were confident enough with their number bonds, or quick mental methods such as repeated doubling, to realise that $5^3 \times 2^4$ or 125×16 could be easily evaluated as 2000. So they often attempted 35×125 etc with a distinct lack of success.	annotate this question if partially correct allow 4 for $70\,000x^4$ www; may also include other terms in expansion. Allow marks even if wrong term selected; mark the coefficient of x^4 may be unsimplified, but do not allow 35 in factorial form unless evaluated later or for all three elements seen together (eg in table) but not multiplied
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	(A) $2A + D = 25$ oe	B1		condone lower-case a and d
		i	$4A + 6D = 250$ oe	B1		
		i	$D = 50,$	B1		
		i	$A = -12.5$ oe	B1		
					Examiner's Comments Most candidates formed the correct equations and went on to solve them successfully.	
		i	(B) $\frac{50}{2}(2 \times \text{their } A + 49 \times \text{their } D)$ [= 60 625] or $\frac{20}{2}(2 \times \text{their } A + 19 \times \text{their } D)$ [= 9250]	M1	or $a = \text{their } A + 20D$	
		i	their " $S_{50} - S_{20}$ "	M1	$S_{30} = \frac{30}{2}(a + l)$ oe with $l =$ their $A + 49D$	$S_{30} = \frac{30}{2}(2 \times \text{their } 987.5 + 29 \times \text{their } 50)$
		i	51 375 cao	A1	Examiner's Comments Many achieved full marks here. Of those who didn't, most candidates scored two marks for $S_{50} - S_{20}$ with their A and D . A few used S_{21} and just scored 1. Other candidates earned the first mark for u_{21} and about half then earned the second mark for a correct formula with $n = 30$. Fortunately hardly any candidates tried to sum all 30 terms individually.	
		ii	$\frac{a(r^2 - 1)}{r - 1} = 25$ or $\frac{a(r^4 - 1)}{r - 1} = 250$	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$a \frac{(r^4 - 1)}{r - 1} = \frac{250}{25} \text{ oe}$ $a \frac{(r^2 - 1)}{(r - 1)}$	M1		allow $a(1 + r)$ as the denominator in the quadruple-decker fraction
		ii	and completion to given result www		at least one correct interim step required	$r^2 = x$ oe may be use
		ii	use of $r^4 - 1 = (r^2 - 1)(r^2 + 1)$ to obtain $r^2 + 1 = 10$ www	M1	or multiplication and rearrangement of quadratic to obtain $r^4 - 10r^2 + 9 = 0$ oe with all three terms on one side	or M1 for valid alternative algebraic approaches eg using $a(1 + r) = 25$ and $ar^2 + ar^3 = ar^2(1 + r) = 225$
		ii	$r = \pm 3$	A1		or B2 for all four values correct, B1 for both r values or both a values or one pair of correct values if second M mark not earned
		ii	$a = 6.25$ or -12.5 oe	A1	or A1 for one correct pair of values of r and a	
					Examiner's Comments Most earned the first mark, but then there was much toil for the second mark, which was often not earned due to wrong working or to leaving too much to the marker's imagination. Faced with solving the given statement, most opted for multiplying by $r^2 - 1$ and were then stumped by the quartic. Careless work led to $r^2 = 10$ or 11 . A good number of candidates who successfully found r neglected to find a . A small number of candidates produced elegant work for full marks.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5			$1 - 20x + 150x^2 - 500x^3 + 625x^4$ as final answer	4	part marks can be awarded for earlier stages if final answer incorrect or not fully simplified: M3 for 4 terms correct or for all coefficients correct except for sign errors or for correct answer seen then further 'simplified' or for all terms correct eg seen in table but not combined (condone eg $+(-20x)$ or $+(-20x$ instead of $-20x)$) M2 for 3 terms correct or for correct expansion seen without correct evaluation of coefficients [if brackets missing in elements such as $(-5x)^2$ there must be evidence from calculation that $25x^2$ has been used] binomial coefficients such as 4C_2 are not sufficient – must show understanding of these symbols by at least partial evaluation; or M1 for 1 4 6 4 1 soi, eg in Pascal's triangle or in expansion where powers of 5 have been ignored	for binomial coefficients, 4C_2 or factorial notation is not sufficient but accept $\frac{4 \times 3 \times 2 \times 1}{2 \times 1 \times 2 \times 1} \text{ oe etc}$ any who multiply out instead of using binomial coeffs: look at their final answer and mark as per main scheme if 3 or more terms are correct, otherwise M0 Examiner's Comments Binomial expansion was done well in comparison with previous years. Most candidates remembered to use the correct coefficients and were comfortable multiplying them with powers of 5. There were not too many arithmetic errors.
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	converging + valid reason	1		eg converges to 0, $r = \frac{1}{2}$, difference between terms decreasing, sum of terms converges to 6, G.P. with $ r < 1$
		ii	neither + valid reason	1		eg divergent oe, A.P., $d = 4$ oe, convergent and periodic ruled out with correct reasons
		iii	periodic + valid reason	1	Examiner's Comments A considerable number of candidates ignored the request to state a reason, and therefore failed to score. Some simply wrote out the first few terms of each sequence and others made comments which were too vague to be credited, such as “decreasing, so converging”. A few lost the mark in one or more parts because there was no statement of “convergent” or “neither” — even if a correct reason had been identified.	eg repeating cycle of terms
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	$2S$ cao	B1	<u>Examiner's Comments</u> Most candidates did not earn this mark: in spite of $\frac{2a}{1-r}$ being commonly seen, candidates were unable to make the connection to "S". Those who did, often left their answer embedded in irrelevant working.	
		ii	$\frac{a}{1-r^2}$	M1	if M0, SC1 for $\frac{1-r}{1-r^2} \times S$ oe	
		ii	$\frac{S}{1+r}$ or $\frac{1}{1+r}S$	A1	<u>Examiner's Comments</u> Approximately three quarters of candidates made the correct initial move of $\frac{a}{1-r^2} \cdot A$ few then recognised that factorising the denominator was relevant, but only a tiny minority went on to earn the second mark.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	11.5, 11 and 10.5 oe	B1		ignore labelling
		i	arithmetic and / or divergent	B1	allow AP ignore references to a, d or n Examiner's Comments Nearly all candidates spotted the algebraic definition and correctly found the required terms. A few lost a mark by calculating the first, second and fourth term, and a few thought it was an inductive definition and substituted u_1 in the formula instead of $n = 2$. The most common description was "arithmetic"; a few candidates also earned the mark with "divergent". However, a significant minority either omitted a description altogether, gave an incorrect answer (most commonly "convergent" or "geometric" and occasionally "periodic") or spoiled their correct answer by hedging their bets: for example, "converging arithmetic" was fairly common.	incorrect embellishments such as converging arithmetic..., diverging geometric... do not score. B0 if a choice is given e.g. AP / GP.
		ii	$n = 30$ identified as number of terms in relevant AP	B1		e.g. $1 + 2 + 3 + \dots + 30$ is not a relevant AP
		ii	$S_{30} = \frac{30}{2}(2 \times 11.5 + (30 - 1) \times -0.5)$	M1	or $S_{30} = \frac{30}{2}(11.5 + -3)$	condone one error in a , d or n but do not condone $l = -\frac{1}{2}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	127.5 oe	A1	allow recovery from slip in working (e.g. omission of minus sign) Examiner's Comments A little over half of candidates scored full marks on this question. A surprising number either specifically identified d as $\frac{1}{2}$, or omitted the minus sign when calculating the sum of the A.P., and ended up with an answer of 562.5. Very few of these candidates had the sense that something must have gone wrong. A few others mistakenly identified a as 12, but were still able to score 2 marks. Some candidates did not use the formula, instead writing out all the terms and calculating the sum directly: as often as not the arithmetic went astray and so only the first mark was earned. Approximately one fifth of candidates made no headway. The sigma notation proved insurmountable for a few, and others used the formula for the sum of a geometric progression or simply attempted to find the nth term. Others confused Σu_n with Σn, and thus failed to score.	SC3 if each term calculated and summed to correct answer or for 127.5 unsupported
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9			$ar = 24$ (i) $\frac{a}{1-r} = 150$ (ii) correct substitution to eliminate one unknown	B1* B1* M1dep*	eg subst. of $a = 150(1-r)$ or $r = \frac{150-a}{150}$ in (i) alternatively, subst. of $a = \frac{24}{r}$ or $r = \frac{24}{a}$ in (ii)	allow $ar^{2-1} = 24$ if M0, B1 for both values of r and B1 for both values of a , or B1 for each pair of correct values NB $150r^2 - 150r + 24 [= 0]$ $a^2 - 150a + 3600 [= 0]$
			$r = 0.8$ or 0.2 $a = 30$ or $a = 120$	A1 A1	or A1 for each correct pair of values ignore incorrect pairing if correct values already correctly attributed	A0 if wrongly attributed A0 if wrongly attributed
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	$21\left(\frac{1}{1+2} + \frac{1}{2+2} + \frac{1}{3+2} + \frac{1}{4+2} + \frac{1}{5+2}\right)$ oe soi	M1	may be implied by correct answer	NB $7 + 5.25 + 4.2 + 3.5 + 3$ M0 if extra terms or terms missing
		i	22.95 or $\frac{459}{20}$ or $22\frac{19}{20}$	A1	Examiner's Comments This was done very well. A few candidates didn't appreciate the meaning of Σ and merely listed the terms. Similarly, a small number of candidates simply added the first and the last terms. Very few resorted to AP or GP formulae.	
		ii	$a + 45$ cao	B1	mark the final answer must be explicitly stated	
		ii	$\frac{10}{2}(a + a + \text{their } 45)$	M1	or $\frac{10}{2}(2a + (10 - 1) \times 5)$	condone wrongly attributed answers

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	5 (2a + 45) or 10a + 225 cao isw	A1	ignore further work attempting to find a Examiner's Comments Most recognised the arithmetic progression, but some were uncomfortable with a non-numerical a and made a spurious attempt to find its value. For a significant number of candidates, the tenth term was either left as $a + 9 \times 5$ or simplified thus: $a + 45 = 45a$. In both cases an easy mark was lost. Many started again to find the sum of the first ten terms, and did so successfully. There was no credit for those candidates who left their answers in terms of a and d . A number of candidates wasted time by trying to find the numerical value of a .	B2 if correct answer derived from adding terms separately
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		i	3×3^7 oe	M1	condone 1×3^7	do not award if only seen in sum of terms of GP
		i	6561	A1	or B2 if unsupported Examiner's Comments Many candidates found this a straightforward question and answered it successfully. 3×3^7 or 3^8 was frequently seen as the method. 1×3^7 was seen occasionally as was an unsupported 2187. Many candidates opted for longer methods, which included lists of powers of 3 and/or diagrams.	if 0, SC1 for 2187 unsupported
		ii	valid attempt to sum a GP with $r = 3$ and $n = 15$	M1	eg $3 + 3^2 + \dots + 3^{15}$	must see at least first two terms and last term NB 7 174 453 implies M1 from $1 + 3 + \dots + 3^{14}$
		ii	$\frac{3(3^{15} - 1)}{3 - 1}$ oe	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	21523359	A1	or B2 if M1M0 or B3 if unsupported Examiner's Comments This was answered successfully by the majority of the candidates, including those who simply worked out the terms of the GP and added them. Stronger candidates achieved a correct answer from a correct formula although there was a good number of unsupported correct answers. A variety of incorrect formulae were in evidence, among the more common were: $\frac{1(3^{15}-1)}{3-1}, \frac{3(3^{14}-1)}{3-1} \text{ and } \frac{3(3^{15}-1)}{15-1}.$ Many candidates listed the terms either evaluated or expressed as powers and summed their list to achieve a correct result.	
		iii	$\frac{3(3^n-1)}{3-1} > 1000000$ oe	M1*		M0 for working backwards
		iii	eg $3^{n+1} > 2000003$ or $3^n > \frac{2000000}{3} + 1$ www			M0 if = or < used
		iii	correctly taking logs of both sides	M1dep*	eg $\log 3^{n+1} > \log 2000003$ www or $\log 3^n + \log 3 > \log 2000003$ www; may be implied by next stage of working	at least one previous progressive interim step needed with no wrong working; M0dep* for $\log(3^n - 1) > \dots$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	eg $(n + 1) \log 3 > \log 2000003$ or $n \log 3 > \log 2000003 - \log 3$			
		iii	eg $n + 1 > \frac{\log 2000003}{\log 3}$ and completion to $n > \frac{\log 2000003}{\log 3} - 1$	A1	without any wrong working	do not allow recovery from bracket errors at any stage

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$n = 13$ seen	B1	B0 for $n \geq 13$ or $n > 13$ Examiner's Comments This part differentiated well between the best and good candidates. Setting up the initial inequality proved beyond many, even those who had successfully used the formula for the sum of a GP in part (ii). There were some splendid examples of well-argued proofs, but getting beyond the first M1 was unusual. Those who started with the $1-3^n$ version of the formula were rarely successful, as basic rules of inequalities when multiplying by a negative were forgotten. Some candidates also thought that $3 \times 3^n = 9^n$. The taking of logs produced problems. Sadly, many candidates who managed to get started with the inequality, didn't answer the last part, whereas weak candidates often went straight for this. A surprising number of candidates thought it perfectly reasonable to have non-integer numbers of generations.	
		iv	valid attempt to sum a GP with $r = 2$ and $n = 15$	M1*	if correct eg $2 + 2^2 + \dots + 2^{15} = 65\,534$	NB 32767 implies M1 from $1 + 2 + \dots + 2^{14}$
		iv		M1dep*	with their $65\,534 < \text{their } 21\,523\,359$	
		iv	their $21\,523\,359 - \text{their } 65\,534$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	21 457 825 isw	A1	allow B3 for 21 457 825 unsupported Examiner's Comments Those candidates who answered parts (i) and (ii) successfully usually went on to achieve full marks in part (iv). A few worked with the fifteenth term, rather than the sum of the first fifteen terms.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13			$u_2 = \frac{10}{2^2}, u_3 = \frac{10}{\text{their } 2.5^2}, u_4 = \frac{10}{\text{their } 1.6^2}$ isw $2 + u_2 + u_3 + u_4$ soi 10.00625 or $\frac{1601}{160}$ or $10\frac{1}{80}$ cao isw	M1* M1dep* A1	NB 2.5, 1.6, 3.90625 or $\frac{10}{4}, \frac{8}{5}, \frac{125}{32}$ must be the sum of 4 terms only B3 if unsupported Examiner's Comments A little under half of candidates achieved full marks on this question. Approximately 20% prematurely rounded their answers and lost the final accuracy mark, and a few found the sum of the second to fifth terms inclusive instead of the first to fourth. The most common error for those who failed to score at all was to treat the sequence as being defined algebraically, but a few candidates misused the formula for the sum of an arithmetic or geometric progression.	may be implied by eg sight of 3.9 and answer of 10.0 NB 2.5, 1.1, 0.625 scores MOMO
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14			$1.5 + (4 - 1)d = 12$ or better	M1	or $1.5 \times r^{(4-1)} = 12$ or better	if first M0 B0 allow B3 for $d = 3.5$ and $r = 2$;
			$d = 3.5$	A1	$r = 2$	B2 for one of these; may be embedded in calculation of difference
			$r = 2$	B1	$d = 3.5$	
			$1.5 \times \text{their } 2^9 - (1.5 + 9 \times \text{their } 3.5)$ oe	M1	M0 for use of their S_{10} in either term	NB $768 - 33$
			differences = 735	A1	Examiner's Comments This was done very well indeed, with many candidates scoring full marks. A few slipped up with the arithmetic and lost the accuracy marks, but the method was very well understood.	allow -735
			Total	5		
15			(i) [5], 10, 5, [10]	M1	ignore extra terms	condone wrongly attributed terms
			[10 + 5 + 10 =] 25	A1	not from wrong working	B2 for 25 unsupported
			(ii) 0	B1	Examiner's Comments Many candidates had difficulty with this question. In some cases it would seem that this was due to a failure to read the question properly, but it was also apparent that a significant minority did not understand how to generate the terms of the sequence. Even many of those who did generate the terms successfully then either ignored the sigma notation or summed an incorrect number of terms.	
			Total	3		

Question		Answer/Indicative content	Marks	Part marks and guidance	
16		$\sqrt[3]{1-2x} = (1-2x)^{1/3}$ $= 1 + \frac{1}{3}(-2x) + \frac{\frac{1}{3}(-\frac{2}{3})}{2!}(-2x)^2 + \frac{\frac{1}{3}(-\frac{2}{3})(-\frac{5}{3})}{3!}(-2x)^3 + \dots$ $= 1 - \frac{2}{3}x - \frac{4}{9}x^2 - \frac{40}{81}x^3 + \dots$ Valid for $-\frac{1}{2} < x < \frac{1}{2}$ or $x < \frac{1}{2}$	B1 M1 B1 B1 B1	$n = 1/3$ only. Do not MR for $n \neq 1/3$ all four correct unsimplified binomial coeffs (not nCr) soi condone absence of brackets only if it is clear from subsequent work that they were assumed $1 - \frac{2}{3}x$ www in this term $\dots - \frac{4}{9}x^2$ www in this term (not if used $2x$ for $(-2x)$ throughout) $\dots - \frac{40}{81}x^3$ www in this term If there is an error in say the third coeff of the expansion then M0 B1B0B1 is possible. Independent of expansion Allow $\leq$'s (valid in this case) or a combination. Condone also, say, $-\frac{1}{2} < x < \frac{1}{2}$ but not $x < \frac{1}{2}$ or $-1 < 2x < 1$ or $-\frac{1}{2} > x > \frac{1}{2}$ Examiner's Comments The method for finding the binomial expansion was understood by almost all candidates. Many candidates scored full marks here. The most common errors were sign errors, the omission of the validity or the use of $2x$ throughout instead of $(-2x)$.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	6	


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17		i	$\frac{x}{(1+x)(1-2x)} = \frac{A}{1+x} + \frac{B}{1-2x}$			
		i	$\Rightarrow x = A(1-2x) + B(1+x)$	M1	expressing in partial fractions of correct form (at any stage) and attempting to use cover up, substitution or equating coefficients Condone a single sign error for M1 only.	
		i	$x = \frac{1}{2} \Rightarrow \frac{1}{2} = B(1 + \frac{1}{2}) \Rightarrow B = \frac{1}{3}$	A1	www cao	
		i	$x = -1 \Rightarrow -1 = 3A \Rightarrow A = -\frac{1}{3}$	A1	www cao	
		i			(accept $A/(1+x) + B/(1-2x)$, $A = -1/3$, $B = 1/3$ as sufficient for full marks without needing to reassemble fractions with numerical numerators) <u>Examiner's Comments</u> Whilst almost all candidates knew the general method for expressing the given fraction in partial fractions, there were a surprising number of numerical errors.	
		ii	$\frac{x}{(1+x)(1-2x)} = \frac{-1/3}{1+x} + \frac{1/3}{1-2x}$			
		ii	$= \frac{1}{3} [(1-2x)^{-1} - (1+x)^{-1}]$			
			$= \frac{1}{3} [1 + (-1)(-2x) + \frac{(-1)(-2)}{2}(-2x)^2 +$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \frac{1}{3}[1 + 2x + 4x^2 + \dots - (1 - x + x^2 + \dots)]$	A1 A1	$1 + 2x + 4x^2$ $1 - x + x^2$ (or $1/3$ $-1/3$ of each expression, ft their A/B) If $k(1 - x + x^2)$ (A1) not clearly stated separately, condone absence of inner brackets (i.e. $1 + 2x + 4x^2 - 1 - x + x^2$) only if subsequently it is clear that brackets were assumed, otherwise A1A0. [i.e. $-1 - x + x^2$ is A0 unless it is followed by the correct answer] Ignore any subsequent incorrect terms	
		ii	$= \frac{1}{3}(3x + 3x^2 + \dots) = x + x^2 + \dots$ so $a = 1$ and $b = 1$	A1	or from expansion of $x(1 - 2x)^{-1}(1 + x)^{-1}$ www cao	
		ii	OR $x(1 - x - 2x^2) = x(1 - (x + 2x^2))$ $= x(1 + x + 2x^2 + (-1)(-2)(x + 2x^2)^2/2 + \dots)$	M1	correct binomial coefficients throughout for $(1 - (x + 2x^2))$ oe (i.e. $1, -1$), at least as far as necessary terms $(1 + x)$ (NB third term of expansion unnecessary and can be ignored)	
		ii	$= x(1 + x + 2x^2 + x^2 \dots)$	A2	$x(1 + x)$ www	
		ii	$= x + x^2 \dots$ so $a = 1$ and $b = 1$	A1	www cao	
		ii	Valid for $-\frac{1}{2} < x < \frac{1}{2}$ or $ x < \frac{1}{2}$	B1	independent of expansion. Must combine as one overall range. condone $\leq$ s (although incorrect) or a combination. Condone also, say $-\frac{1}{2} < x < \frac{1}{2}$ but not $x < \frac{1}{2}$ or $-1 < 2x < 1$ or $-\frac{1}{2} > x > \frac{1}{2}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			Examiner's Comments Most candidates were able to use the binomial expansion correctly although there were sign errors - often from using $(-2x)$ as $(2x)$. The most common error - which was very common - was using $\frac{1}{3(1+x)} = 3(1+x)^{-1}$ and similarly for $\frac{1}{3(1-2x)}$. The other frequent error was in the validity. Some candidates omitted this completely but many others failed to combine the validities from the two expansions, or failed to choose the more restrictive option.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18			$(4 + x)^{\frac{3}{2}} = 4^{\frac{3}{2}} \left(1 + \frac{1}{4}x\right)^{\frac{3}{2}}$ $= 8 \left(1 + \frac{3}{2} \cdot \frac{1}{4}x + \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2!} \left(\frac{1}{4}x\right)^2 + \dots\right)$ $= 8 + 3x$ $+ 3/16 x^2$	M1 M1 A1 A1	dealing with the '4'to obtain $4^{3/2} \left(1 + \frac{x}{4}\right)^{3/2}$ $4^{3/2} = 2^3 = 8$ (and having all the powers of 4 correct) correct binomial coeffs for n = 3/2 ie 1, 3/2, 3/2.1/2.1/2! Not nCr form Indep of coeff of x Indep of first M1 8 + 3x www ... + 3/16 x ² www Ignore subsequent terms	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Valid for $-4 < x < 4$ or $ x < 4$	B1	accept $\leq$ s or a combination of $<$ and $\leq$, but not $-4 > x > 4$, $x > 4$, or say $-4 < x$ condone $-4 < x < 4$ Indep of all other marks Allow MR throughout this question for $n = m/2$ where $m \in \mathbb{N}$, and m odd and then -1 MR provided it is at least as difficult as the original. Examiner's Comments Much here depended upon the candidate's ability to factorise correctly. On too many occasions the factor was found to be 4 or $\frac{1}{4}$ instead of 8. The general method for expanding the binomial expansion was understood and the binomial coefficients were usually correct. Some who had factorised correctly then forgot to include the 8 at the final stage. The validity was often correct but was sometimes omitted and a variety of incorrect responses were also seen including $-\frac{1}{4} < x < \frac{1}{4}$. Good candidates scored well in this question.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	$\frac{1}{\sqrt[3]{1-2x}} = (1-2x)^{-1/3}$	B1	$n = -1/3$. See below SC for those with $n = 1/3$	
		i	$= 1 + \left(-\frac{1}{3}\right)(-2x) + \frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)}{2!}(-2x)^2 + \dots$	M1	All three correct unsimplified binomial coefficients (not nCr) so condone absence of brackets only if it is clear from subsequent work that they were assumed	
		i		B1	$1 + (2/3)x + \dots$ www	
		i	$= 1 + \frac{2}{3}x + \frac{8}{9}x^2 + \dots$	B1	$(8/9)x^2$ www in this term If there is an error, in say, the third coefficient of the expansion then M0B1B0 is possible SC For $n = 1/3$ award B1 for $1 - (2/3)x$ and B1 for $-(4/9)x^2$ (so max 2 out of the first 4 marks)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	Valid for $-\frac{1}{2} < x < \frac{1}{2}$ or $ x < \frac{1}{2}$	B1	Independent of expansion. Accept, say, $-1/2 < x < 1/2$ or $-1/2 \leq x < 1/2$ (must be strict inequality for $+1/2$) Examiner's Comments The most common mistake in part (i) was to use a value of 2 rather than -2 as the coefficient of x in each term of the expansion. The binomial coefficients were nearly always correct though a small number missed the $2!$ from the denominator of the x^2 term. While the majority of candidates used the correct value of n a small minority incorrectly used $n = \frac{1}{3}$ or $-\frac{2}{3}$. The range of validity of the expansion was done much better than in previous years although the most common mistake was to give non-strict inequalities. Other mistakes included: $-\frac{1}{2} < -x < \frac{1}{2}$ $x > \frac{1}{2}$ $\frac{1}{2} < x < -\frac{1}{2}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{1-3x}{\sqrt[3]{1-2x}} = (1-3x)(1 + \frac{2}{3}x + \frac{8}{9}x^2 + \dots)$		Examiner's Comments In part (ii) the majority of candidates correctly multiplied their answer from part (i) with $(1-3x)$ and simplified this expression correctly to obtain the correct values of a and b. It was concerning, however, that a number of candidates wrote $(1-3x)(1 + \frac{2}{3}x + \frac{8}{9}x^2 + \dots) = 1 + \frac{2}{3}x + \frac{8}{9}x^2 - 3x - 2x - \frac{8}{3}x^2 + \dots$ or even more worryingly expanded $(1-3x)^1$ as $1-3x +$ higher order terms in x.	
		ii	$= 1 + \frac{2}{3}x + \frac{8}{9}x^2 - 3x - 2x^2 + \dots$	M1		
		ii		A1ft		
		ii	$= 1 - \frac{7}{3}x - \frac{10}{9}x^2 + \dots$	A1		
			Total	8		

Question		Answer/Indicative content	Marks	Part marks and guidance	
20		$\left(1 + \frac{x}{p}\right)^q = 1 + q\frac{x}{p} + \frac{q(q-1)}{2!}\left(\frac{x}{p}\right)^2 + \dots$ $\frac{q}{p} = -1 \frac{q(q-1)}{2p^2} = \frac{3}{4}$ $q = -p \Rightarrow \frac{-p(-p-1)}{2p^2} = \frac{3}{4} \text{ or } \frac{q(q-1)}{2q^2} = \frac{3}{4}$ $\Rightarrow p = 2$ $\Rightarrow q = -2$	M1* A1 A1 M1dep* A1 A1ft	One of $\frac{q}{p}x \text{ or } \frac{q(q-1)}{2!}\left(\frac{x}{p}\right)^2 \text{ (soi),}$ for example, $\frac{q}{p} = -1$ scores M1 A1 Allow x's on both sides of equations (if correct) Eliminating p (or q) from simultaneous equations (not involving x) involving both variables oe – if M1A1A1 awarded followed by either p or q correct (www) this implies this M mark $p = 2$ www (or $q = -2$) $q = -2$ (or $p = 2$) for second value, ft their p or q (e.g. the negative of their p or q) provided first 4 marks awarded and only a single computational error in the method – so must be a correct method for solving their equation in p or q (ignore mention of p and/or $q = 0$)	

Question	Answer/Indicative content	Marks	Part marks and guidance
	Valid for $\left \frac{x}{2}\right < 1 \Rightarrow x < 2$	A1	or $-2 < x < 2$ www, allow $-2 < x < 2$ but not say, $x < 2$ SC If M0 M0 awarded and no wrong working seen then B1 for $p = 2$ and $q = -2$, B1 for $-2 < x < 2$ (oe) so max 2 marks Guidance for solving quadratics on this paper: use of correct quadratic equation formula (if formula is quoted correctly then only one sign slip is permitted, if the formula is quoted incorrectly M0, if not quoted at all substitution must be completely correct to earn the M1) or factorising (giving their x^2 term and one other term when factors multiplied out) or completing the square (must get to the square root stage involving $\pm$ and arithmetical errors may be condoned provided that perfect square term was correct) Examiner's Comments The binomial expansion of $\left(1 + \frac{x}{p}\right)^q$ was done extremely well by the vast majority of candidates with the most common error being the failure to correctly deal with the x^2 term with many giving the coefficient (of this term) as $\frac{q(q-1)}{2p}$ rather than the

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					correct $\frac{q(q-1)}{2p^2}$. It was surprising how few candidates could go on to form the correct pair of simultaneous equations and fewer still who could solve this pair of equations accurately and successfully. Those candidates who correctly found the value of p usually went on to state the set of values of x for which the expansion was valid.	
			Total	7		
21			-15	B3(AO1.1) (AO1.1) (AO1.1) [3]	B2 for 15 or $5 \times (-3)^1$ or better OR B1 for 5 or 1 5 10 10 5 1 row of Pascal's triangle seen	Do not accept ${}_5C_4$ as a correct element without evaluation to 5
			Total	3		
22			$\frac{3}{1-r} = 8$ $\Rightarrow 3 = 8(1-r)$ $\Rightarrow r = \frac{5}{8}$	M1(AO1.1) M1(AO1.1) A1(AO1.1) [3]	Use of correct formula Clearing fraction	
			Total	3		

$\approx 1 + \frac{1}{2}(-2x) + \frac{\frac{1}{2}\left(\frac{-1}{2}\right)}{2!}(-2x)^2 + \frac{\frac{1}{2}\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)}{3!}(-2x)^3$ $= 1 - x - \frac{1}{2}x^2 - \frac{1}{2}x^3$	[4]	correct terms or $x < \frac{1}{2}$	series converges for $x = \pm \frac{1}{2}$ candidates are not expected to know this but allow $\leq$ for either or both inequalities.
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Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	4	

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Question			Answer/Indicative content	Marks	Part marks and guidance		
24			$\sum_{r=1}^4 \ln \frac{r}{r+1} = \ln \frac{1}{2} + \ln \frac{2}{3} + \ln \frac{3}{4} + \ln \frac{4}{5}$ $= \ln 1 - \ln 2 + \ln 2 - \ln 3 + \ln 3 - \ln 4 + \ln 4 - \ln 5$ $= 0 - \ln 2 + \ln 2 - \ln 3 + \ln 3 - \ln 4 + \ln 4 - \ln 5$ $= -\ln 5$	B1(AO1.1) M1(AO3.1a) E1(AO2.2a) [3]	soi use of $\ln \left(\frac{a}{b} \right) = \ln a - \ln b$ or $\ln a + \ln b = \ln ab$ $\ln \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \right) = \ln \frac{1}{5}$ $= \ln 1 - \ln 5$ $= -\ln 5$ AG		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
25		a	(i) $f'(x) = e^{-x} \cos x - e^{-x} \sin x$ $f'(x) = 0$ and $e^{-x} \neq 0 \Rightarrow \cos x = \sin x$ $\Rightarrow \tan x = 1$ $\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$	M1(AO3.1a) A1(AO1.1) E1(AO2.2a) M1(AO1.1) A1(AO1.1)	product rule correct Use of $\frac{\sin}{\cos} = \tan$ $x = \frac{\pi}{4}$ (condone 45°) $\frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4},$...		
		a	So an AP with $d = \pi$ (ii) $y = \frac{\sqrt{2}}{2}e^{-\frac{\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{5\pi}{4}}, \frac{\sqrt{2}}{2}e^{-\frac{9\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{13\pi}{4}}$) This is a GP with $r = -e^{-\pi}$	E1FT(AO2.1) M1(AO3.10a) A1(AO1.1) E1FT(AO2.1) [9]	must state the common difference substituting one value of x into $f(x)$ must state common ratio, www	FT their values of x FT their values of y	
		b	Yes with explanation that values of x would continue to be separated by π and so values of y would continue to have same common ratio.	E1(AO2.2a) [1]			
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
26			$(2x)^5 + 5(2x)^4(-3) + 10(2x)^3(-3)^2 + 10(2x)^2(-3)^3$ $+5(2x)(-3)^4 + (-3)^5$ $32x^5 - 240x^4 + 720x^3 - 1080x^2 + 810x - 243$	M1(AO2.1) M1(AO1.1) A1(AO1.1) A1(AO1.1) [4]	Binomial coefficients 6 terms in powers of x from 0 to 5 Five terms correct Six terms correct	May be unsimplified eg 5C_2 or 1 5 10 10 5 1 seen From 5 to 0	
			Total	4			
27			Sequence of the form $\ln a$, $\ln ar$, $\ln ar^2 \dots$ $\ln a$, $\ln a + \ln r$, $\ln a + 2\ln r \dots$ Arithmetic sequence with common difference $\ln r$	M1(AO 2.5) M1(AO 3.1a) E1(AO 2.2a) [3]	Use of notation for geometric sequence Laws of indices used Laws of indices used		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
28		a	$(1-2x)^{-\frac{1}{2}} \approx 1 + \left(-\frac{1}{2}\right)(-2x) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-2x)^2 + \dots$ $= 1 + x + \frac{3}{2}x^2 \quad \text{AG}$	M1(AO 1.1b) A1(AO 2.1) [2]	Correct form required; allow sign errors		
		b	Valid when $ x < \frac{1}{2}$	E1(AO 2.3) [1]			
		c	$(1+2x)^{\frac{1}{2}} = 1 + x - \frac{1}{2}x^2 + \dots$ $\left(1 + x - \frac{1}{2}x^2\right)\left(1 + x + \frac{3}{2}x^2\right)$ $= 1 + 2x + 2x^2$ Alternative method $\sqrt{\frac{1+2x}{1-2x}} = \sqrt{\frac{(1+2x)(1+2x)}{(1-2x)(1+2x)}} = \frac{1+2x}{\sqrt{1-4x^2}}$ $= (1+2x)\left(1 + \left(-\frac{1}{2}\right)(-4x^2) + \dots\right)$ $= 1 + 2x + 2x^2$	B1(AO 1.1a) M1(AO 1.1a) A1(AO 1.1b) Alternative method M1 M1 A1 [3]	Product of their expansions attempted		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	$\sqrt{\frac{1.1}{0.9}} = \frac{\sqrt{11}}{3} \approx 1 + 2 \times \frac{1}{20} + 2 \times \left(\frac{1}{20}\right)^2$ $\sqrt{11} \approx 3.315$	M1(AO 2.1) A1(AO 2.2a) [2]	Obtaining an expression involving $\sqrt{11}$ Or $\frac{663}{200}$		
			Total	8			
29			Gradient of chord $= \frac{(x+h)^3 - x^3}{h} \text{ oe}$ $= \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$ Derivative is limit of gradient of chord as $h \rightarrow 0$ $3x^2 + 3xh + h^2 \rightarrow 3x^2 \text{ AG}$	M1(AO 2.1) M1(AO 1.1a) A1(AO 1.1) E1(AO 2.4) B1(AO 2.2a) [5]	Expansion of cubic attempted, with at least some terms correct Completely correct expansion	Not just $x^3 + h^3$	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
30			DR $\frac{b^2 - 13}{1 - \frac{1}{b}} = -6$ $\frac{b(b^2 - 13)}{b - 1} = -6$ $b(b^2 - 13) = -6(b - 1)$ $b^3 - 7b - 6 = 0$ $(-1)^3 - 7 \times (-1) - 6 = 0$ $(b + 1)(b^2 - b - 6) = 0$ $(b + 1)(b + 2)(b - 3) = 0$ Roots $-1, -2, 3$ -1 cannot be the common ratio of a geometric sequence with a sum to infinity Possible common ratios are $-\frac{1}{2}$ and $\frac{1}{3}$	M1(AO 3.1a) M1(AO 3.1a) A1(AO 1.1) M1(AO 1.1) M1(AO 2.1) A1(AO 1.1) M1(AO 1.1) B1(AO 2.3) A1FT(AO 3.2a) [9]	Use of sum to infinity to form an equation Starting to clear fractions Correct equation without fractions Cubic with zero on one side Use of factor theorem to search for factor Correct fact Method for solving quadratic Rejection of root that does not make sense in the context Follow through their values of b		
			Total	9			

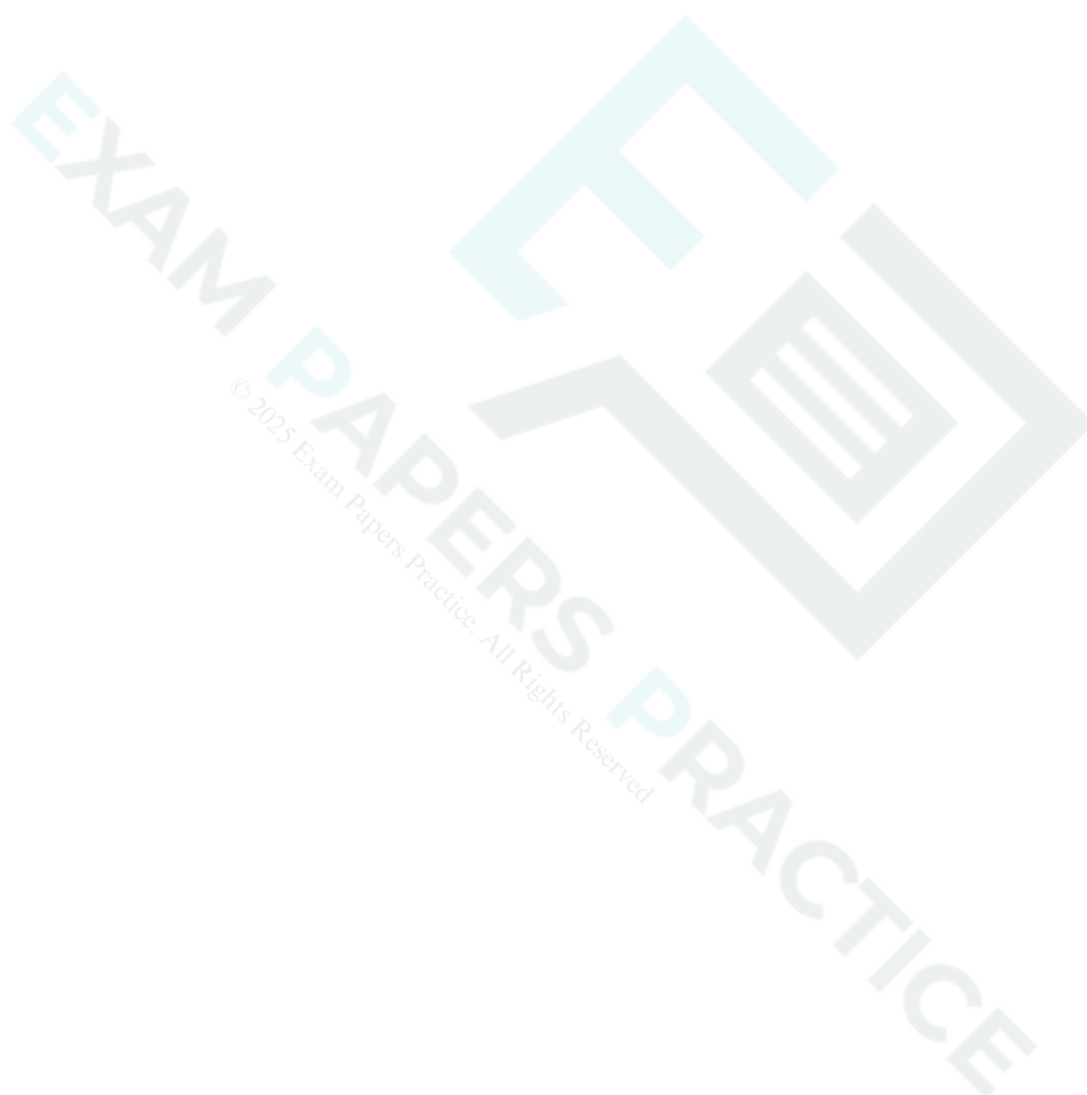
Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i	$\binom{50}{4} = \frac{50!}{4!46!} = 230300$	M1 A1 [2]	Examiner's Comments This part was answered well by the majority of the candidates. A small minority of the candidates used permutations instead of combinations and a very few simply gave an answer of $4! = 24$.	
		ii	$\frac{17}{50} \times \frac{16}{49} \times \frac{15}{48} \times \frac{14}{47} = \frac{17}{1645} = 0.0103$ Or: $\binom{17}{4} \div \binom{50}{4} = \frac{2380}{230300} = 0.0103$	M1 A1 [2] M1 A1	$17/50 \times$ or $0.34 \times$ NB $\left(\frac{17}{50}\right)^4$ or 0.34^4 scores M1A0 But M0 if part of a binomial expression CAO Uncancelled fraction gets M1A0 Uncancelled fraction gets M1A0 Allow 0.010 with working but not 0.01 Examiner's Comments This was answered well by many candidates. However a common misconception was to assume 'with replacement' probability giving an answer of $0.34^4 = 0.0134$. A few candidates gave their answer in fractional form but failed to cancel, hence losing the accuracy mark.	

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Question		Answer/Indicative content	Marks	Part marks and guidance		
32		$a^5 = 32$	B1	must have evidence that they have considered the constant term	NB examiners must use annotation in this part; a tick where each mark is earned is sufficient	
		$a = 2$	B1			
		$10a^2b^3 [= -1080]$	B1	B0 for $a = \pm 2$, but allow them to gain all marks for b if earned	B0 for eg $10a^2bx^3 = -1080x^3$	
		$4b^3 = -108$ oe	B1	may include x^3 on both sides, or $(bx)^3$ on left and x^3 on right; may have subst their a^2 ; condone poor notation with inconsistent xs.		
		$b = -3$	B1	for subst $a = 2$ in $10a^2b^3 = -1080$ oe	B0 for $4b^3 = -108x^3$ etc	
			[5]	if 0 in qn, allow B1 for 1 5 10 10 5 1 row of Pascal's triangle seen or for ${}^5C_3 = 10$	those trialling factors of -108: Allow up to 3 marks (B0,B1,B1 if earned,B0,B1) for reaching $a = 2$ and $b = -3$ with trialling unless explicit reference to 32 in checking, in which	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					case award up to full marks (in effect explicit reference showing their solution fits both constraints triggers 1st and 4th B1s) Examiner's Comments This problem-solving binomial expansion question discriminated extremely well. Some candidates misunderstood the concept of the constant term being 32 and this was then applied incorrectly in a variety of ways, either being assigned to the value of a or to 5C_3. Another common error was to work with the term bx^3 rather than $(bx)^3$ sometimes leading to an answer $b = 27$ or -27. Having x's on only one side of an equation and then ignoring them until the last statement was also common, as was a correct $b^3 = -27$ followed by the loss of the negative sign, leading to $b = 3$. Candidates' trialling factors of -108 (with no consideration of the 32) often reached correct values for a and b but were not awarded full marks since this went against the rubric on the front cover which requires that candidates show sufficient	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					detail of the working to indicate that a correct method has been used. However even the poorest candidates usually gained a mark for identifying the binomial coefficient 10.	
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	$3 \times 1 + 2 + 3 \times 2 + 2 + 3 \times 3 + 2$ $+ 3 \times 4 + 2 + 3 \times 5 + 2$ oe soi 55	B1 B1 [2]	or $3 \times \frac{1}{2} \times 5 \times (5+1) + 2 \times 5$ or $\frac{5}{2} [2 \times 5 + (5-1) \times 3]$ B2 for 55 unsupported Examiner's Comments This was done very well. A small minority of candidates failed to score, usually through misusing formulae associated with arithmetic or geometric progressions. A small number of candidates demonstrated the correct method, but slipped up with arithmetic.	
		ii	$4.2 + 5d = 1.8$ soi -0.48 or $-\frac{12}{25}$ isw	M1 A1 [2]	or $(1.8 - 4.2) \div 5$ oe M0 for $(4.2 - 1.8) \div 5$ if not recovered B2 for correct answer unsupported Examiner's Comments This was done very well, too. However, some candidates failed to appreciate that d had to be negative, and a few interchanged a and d .	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
34		i	[year 10] A : 39 000 B : 38 783.205...isw r.o.t. to 6 or more significant figures [year 11] A : 40 000 B : 40 722.365...isw r.o.t. to 6 or more significant figures	B1 B1 B1 B1 [4]	B0 for any which are wrongly attributed or 38 800 or 38 780 or 38 783 or 40 700 or 40 720 or 40 722 Examiner's Comments The majority of candidates gained full marks on this question. A few candidates listed all the terms and lost accuracy on the way, and a few misused the formulae.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	A: $\frac{17}{2}(2 \times 30000 + 16 \times 1000)$ or $\frac{17}{2}(30000 + 46000)$ = 646 000 B: $\frac{25000(1.05^{17} - 1)}{1.05 - 1}$ = 646 009.15...r.o.t. to 6 significant figures or more	M1 A1 M1 A1 [4]	if M0 and B0 allow SC1 for 30 000 + 31 000 + ... + 46 000 = 646 000 if M0 and B0 allow SC1 for 25 000 + 25 000 × 1.05 + ... + 25 000 × 1.05¹⁶ = 646 009.15... 646009... unsupported is M0A0 A0 for 646 000 only after award of M1	if M0 then B2 for complete sum written out and correct answer obtained if M0 then B2 for complete sum written out and correct answer obtained	

Examiner's Comments

This part of the question was also very well done, but some candidates did not give enough detail to "show that" Arif and Bettina earned the same amount to the nearest £100. A common mistake was to write down Bettina's earnings as £646 000 without showing the value before rounding.

$0\ 000 \times 1.05^n) >$ $0\ 000) \text{ oe}$ $05^n) > \log_{10}(M + 500$ $\log_{10} 500\ 000 \text{ oe}$ $(+ 500000) - \log_{10} 500000$ $\log_{10} 1.05$	A1 B1 M1 A1
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Question			Answer/Indicative content	Marks	Part marks and guidance		
					e step	B0 for $n > 26$	
					NB $n > 25.08...$		
					Examiner's Comments A minority of candidates presented clear, concise solutions to derive the inequality, and went on to obtain the correct value of n . Many candidates, however, did not attempt the derivation or started with the final statement. A few went on to obtain the correct value of n , although 25 was a common wrong answer.		
			Total	13			

Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	$\frac{5-x}{(2-x)(1+x)} = \frac{A}{2-x} + \frac{B}{1+x}$ $\Rightarrow 5-x = A(1+x) + B(2-x)$ $x=2 \Rightarrow 3=3A, A=1$ $x=-1 \Rightarrow 6=3B \Rightarrow B=2$	M1 A1 A1 [3]	Cover up, substitution or equating coefficients Examiner's Comments Part (i) was answered extremely well with the vast majority of candidates correctly expressing $\frac{5-x}{(2-x)(1+x)}$ in partial fractions.	
		ii	$\frac{A}{2-x} = \frac{A}{2} \left(1 - \frac{1}{2}x\right)^{-1}$ $= \frac{A}{2} \left(1 + (-1)\left(-\frac{1}{2}x\right) + \frac{(-1)(-2)}{2!} \left(-\frac{1}{2}x\right)^2 + \dots\right)$ $= A \left(\frac{1}{2} + \frac{1}{4}x + \frac{1}{8}x^2 + \dots\right)$ $\frac{B}{1+x} = B(1+x)^{-1} = B(1-x+x^2+\dots)$ $\frac{5-x}{(2-x)(1+x)} = \frac{5}{2} - \frac{7}{4}x + \frac{17}{8}x^2 + \dots$	B1 M1 A1ft A1ft A1 [5]	Or equivalent All three correct unsimplified binomial coefficients (not nCr) so for either expansion i.e. 1, -1 and $\frac{(-1)(-2)}{2}$. Or correct simplified coefficients seen Ignore any subsequent incorrect terms – ft their A from (i) only Ignore any subsequent incorrect terms – ft their B from (i) only www cao – ignore any higher order terms stated – isw after correct expansion seen Examiner's Comments	

Question			Answer/Indicative content	Marks	Part marks and guidance
					In part (ii) most candidates used their answer to part (i) in their attempt to find the binomial expansion of $\frac{5-x}{(2-x)(1+x)}$ although some candidates did (with varying degrees of success) attempt to expand $(5-x)(2-x)(1+x)^{-1}$ directly. Whilst the majority of candidates correctly dealt with the expansion of $\frac{2}{1+x}$ (and so scored at least two marks in this part) it was surprising how many candidates (at this level) struggled in re-writing $\frac{1}{2-x}$ as $\frac{1}{2}\left(1-\frac{1}{2}x\right)^{-1}$. In some cases it was clear that candidates either did not realise or even recognise that the 2 inside the bracket had to be removed before this term could be binomially expanded. Those candidates who expanded both terms correctly usually went on to score full marks.
			Total	8	

Question			Answer/Indicative content	Marks	Part marks and guidance	
36			$na = 6$ $\frac{n(n-1)}{2!}a^2 = -6$ Substituti ^{a = 6/n} in second on of equation oe $18(n-1) = -6n$ soi $n = 3/4$ $a = 8$	M1(AO3. 1a) A1(AO2. 1) M1(AO1. 1b) A1(AO1. 1b) A1(AO1. 1b) A1(AO1. 1b) [6]		
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance		
37		a	$n = 500$ $\frac{500}{2} \times 1002$ 250 500	B1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Use of formula for sum of AP		
		b	(i) $\frac{n}{2}(2 + 2n) < 110$ So $n + n^2 < 110$ (ii) $n^2 + n - 110 < 0$ $\Rightarrow (n + 11)(n - 10) < 0$ $[-11 <] x < 10$ {1, 2, 3, 4, 5, 6, 7, 8, 9} oe	E1(AO2.4) [1] M1(AO1.1a) M1(AO1.1) A1(AO1.1) [3]	AG Or roots of quadratic are 10 and -11 Or relevant sketch of quadratic Correct set of numbers clearly stated	At least one line of working leading to given answer is required	
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
38		a	$x^3 = \left(y - \frac{m}{3y}\right)^3$ $= y^3 - 3y^2\left(\frac{m}{3y}\right) + 3y\left(\frac{m}{3y}\right)^2 - \left(\frac{m}{3y}\right)^3$ $= y^3 - my + \frac{m^2}{3y} - \frac{m^3}{27y^3}$ $x^3 + mx = y^3 - \frac{m^3}{27y^3}$ $y^3 - \frac{m^3}{27y^3} = n \Rightarrow (y^3)^2 - \frac{m^3}{27} = ny^3$ This is a quadratic in y^3	M1(AO1.1a) A1(AO1.1) E1(AO2.1) [3]	Successful completion		
		b	$(y^3)^2 - ny^3 - \frac{m^3}{27} = 0$ has distinct real roots when $(-n)^2 - 4\left(-\frac{m^3}{27}\right) > 0$ $n^2 + \frac{4m^3}{27} > 0$ certainly true when $m > 0$	M1(AO3.1a) E1(AO2.1) [2]	Use of their $b^2 - 4ac$		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
39		a	Arithmetic sequence with $a = 50$, $d = 20$ $S_{24} = \frac{24}{2}(2 \times 50 + (24 - 1)20)$ $= £6720$	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Using appropriate formula for sum of an arithmetic sequence with $a = 50$, $d = 20$ Allow full credit for any correct method	Allow for total written out in full	

Examiner's Comments

Some candidates used a brute force method, writing out the complete list of monthly payments and adding them. Most successfully identified this as an arithmetic series and used the correct formula to find the sum of 24 terms.

Partial credit was not awarded where a candidate found the 24th term but did not then attempt to find the sum total.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	Each month is 12% more than the previous, so multiplied by 1.12 giving a geometric sequence with $a = 50$, $r = 1.12$	E1 (AO 2.4) [1]	Clear argument must include the value 1.12 Examiner's Comments Many candidates realised that the 12% increase could be achieved by multiplying by 1.12 which leads to a geometric series. The value 1.12 had to be seen in part (b) for the mark to be credited. Exemplar 2 <i>Because the amount saved is going up in a common ratio, each time it is not the same difference each time.</i> (A1)(ft) Some candidates were not credited the mark as their answer was too vague, as in the exemplar above.  AfL Even if 1.12 was used in part (c) this mark could not be credited if 1.12 was not seen in part (b).	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	Geometric sequence with $a = 50$, $r = 1.12$ $S_{24} = \frac{50(1.12^{24} - 1)}{0.12}$ $= £5907.76$ which is less than Aleela	M1 (AO 3.1a) A1 (AO 1.1b) E1 (AO 2.1) [3]	Using appropriate formula for sum of a geometric sequence with $a = 50$, $r = 1.12$ Allow any suitable rounding FT their values (dep on earning the M marks in part (a) and (c))	Allow for total written out in full	
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
40		a	$f'(x) = \frac{1}{3}(27 - 8x^3)^{-\frac{2}{3}} \times (-24x^2)$ $= \frac{-8x^2}{(27 - 8x^3)^{\frac{2}{3}}}$ $f'(1.5) = -\frac{8 \times 1.5^2}{0}$ and dividing by zero gives the error.	M1 (AO1.1a) A1(AO1.1) E1 (AO2.4) [3]	Using the chain rule Allow unsimplified Sufficient to say “can’t divide by zero” oe <u>Examiner’s Comments</u> This was well answered as most candidates used the chain rule successfully and realised that substituting $x = 1.5$ gives a zero in the denominator		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$(27 - 8x^3)^{\frac{1}{3}} = 27^{\frac{1}{3}} \left(1 - \frac{8}{27}x^3 \right)^{\frac{1}{3}}$ $= 3 \left(1 + \left(\frac{1}{3} \right) \left(-\frac{8x^3}{27} \right) + \frac{\left(\frac{1}{3} \right) \left(-\frac{2}{3} \right)}{2!} \left(-\frac{8x^3}{27} \right)^2 + \dots \right)$ $= 3 - \frac{8x^3}{27} - \frac{64x^6}{2187} + \dots$	B1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) [1]	Dealing with the 27 correctly Using the Binomial expansion substantially correctly Cao Examiner's Comments Many candidates dealt successfully with the 27, but when that was done without clear working shown, could cost 2 marks here. Some candidates simplified their coefficients early and incorrectly, so it was not always clear that the binomial expansion had been used, costing the method mark.  AfL Make your method clear by writing down the factors of each term before simplifying.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	The binomial expansion is $\left -8 \frac{x^3}{27} \right < 1$ valid for $x < 1.5$ and the limits of the integral are completely in this interval.	B1 (AO 2.4) E1 (AO 2.3) [2]	Allow unsimplified but must use correct modulus notation or equivalent Must indicate that the limits of the integral lie in their interval for which the expansion is valid. <u>Examiner's Comments</u> One of the assessment objectives in the specification is to test the ability of a candidate to assess the validity of an argument as in this question. Not many candidates realised that the key to this explanation was to find the range of values for which the binomial expansion is valid. The limits lie well within the valid range so the method is valid.		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	$\frac{0.25}{2}(3 + 2.6684 + 2(2.9954 + 2.9625 + 2.8694))$ $= \frac{0.25}{2} \times 23.3224 = 2.9153$	B1 (AO 1.1a) M1 (AO 1.1b) A1 (AO 1.1b) [3]	$h = 0.25$ used For sum in the bracket – condone one slip. Allow for 2.92 or better	Values from candidates own calculators may differ in the last decimal place.	
		e	There is area between the curve and the top of the trapezia, so some area is missing from the estimate.	E1 (AO 2.4) [1]	Allow for any sensible explanation eg the trapezia are under the curve	“The curve is concave downwards ” on its own is not quite enough	
					Examiner's Comments Most candidates were able to explain this clearly. Some had learned that a curve being concave downwards would give an underestimate but gave no indication as to why that would be, so lost the mark.  AfL Clear annotated sketches can support a mathematical explanation better than an extended written response.		
			Total	12			

Question			Answer/Indicative content	Marks	Part marks and guidance		
41			${}^{15}C_5 (x^2)^5 \left(\frac{1}{x}\right)^{10}$ 3003	M1 (AO 3.1a) A1 (AO 1.1) [2]	Identifying term $(x^2)^5 \left(\frac{1}{x}\right)^{10}$ h	Must see	
			Total	2	Examiner's Comments This was a standard question for which many fully correct solutions were provided. Whilst not required, a clear justification of why ${}^{15}C_5$ is needed may help ensure the correct answer of 3003 is obtained, and could gain partial credit if a minor error is made.		

Question			Answer/Indicative content	Marks	Part marks and guidance		
42			Let the $\frac{c}{r}$, c, cr terms be The geometric mean of first and last is $\sqrt{\frac{c}{r} \cdot cr}$ $\sqrt{\frac{c}{r} \cdot cr} = \sqrt{c^2} = c$; this is the middle term	B1 (AO 3.1a) M1 (AO 1.1) E1 (AO 2.1) [3]	Expressions for three consecutive terms of a GP (any correct form) Expression for GM of first and last term (any correct form) FT their terms AG Correct completion Examiner's Comments Most candidates were able to give general expressions for three consecutive terms of a geometric series and their response progressed successfully. However, some candidates lost a mark by not correctly completing their explanation by relating to the middle term.		
			Total	3			

Question		Answer/Indicative content	Marks	Part marks and guidance		
43		EITHER				
		$(3)^3 + 3(3)^2(-2x) + 3(3)(-2x)^2 + (-2x)^3$ $= 27 - 54x + 36x^2 - 8x^3$ OR $(3 - 2)^2 = (9 - 12x + 4x^2)$ $(3 - 2x)(9 - 12x + 4x^2)$ $= 27 - 54x + 36x^2 - 8x^3$	M1 (AO1.1a) M1 (AO1.1b) A1 (AO1.1b) A1 (AO1.1b) [4] M1 M1 A1 A1 [4]	Use of Binomial coefficients Powers of 3 and $(-2x)$ Condone no brackets or $(2x)$ used. At least 3 simplified terms correct All correct and simplified Attempting to square Multiplying their answer by third bracket At least 3 simplified terms correct All correct and simplified		
				Examiner's Comments This was well answered by the majority of candidates although a significant number did not correctly use brackets round the		

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					($-2x$) and so lost a method mark and the accuracy marks. Some also made the mistake of writing $3^3 = 9$.  Check numerical answers with a calculator.	
			Total	4		
44			${}_5C_3(2)^2x^3$ $40x^3$	M1 (AO 1.1) A1 (AO 1.1) [2]	For ${}_5C_3(2)^2$ or correct coefficient	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance		
45			$a + 2d = 6$ $a + 8d = 30$ $d = 4, a = -2$ $\frac{100}{2}(2 \times (-2) + 99 \times 4)$ 19600	B1 (AO 1.1) M1 (AO 3.1a) M1 (AO 1.1) A1 (AO 1.1) [4]	One equation (could be $6d = 24$) BC Use of formula		
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
46		a	periodic since 1, 0, 1, 0, repeats	B1 (AO1.2) [1]		
		b	convergent, since difference between terms is decreasing	B1 (AO2.4) [1]		
		c	divergent, since magnitude of difference between terms increasing	B1 (AO1.2) [1]	allow "difference between terms increasing"	
			Total	3		
47		a	$1 + (-2)\frac{x}{2} + \frac{(-2)(-3)}{2!}\left(\frac{x}{2}\right)^2 + \frac{(-2)(-3)(-4)}{3!}\left(\frac{x}{2}\right)^3$ $1 - x + \frac{3}{4}x^2 - \frac{1}{2}x^3$	M1 (AO1.1a) A1 (AO1.1) A1 (AO1.1) [3]	allow sign errors and one coefficient error three out of four terms correct all four terms correct	ignore extra terms
		b	$-2 < x < 2$ oe	B1 (AO2.3) [1]		
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
48			$a + 4d = 11$ and $a + 10d = 20$ Equations solved simultaneously to obtain values for a and d . $a = 5$ and $d = 1.5$ $\frac{40}{2}(2 \times 5 + (40 - 1) \times 1.5)$ 1370	B1(AO 1.1a) M1(AO 3.1a) A1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [5]	FT their equations FT their a and d cao		
			Total	5			
49		a	$1 + (-1)(3x) + (-1)(-2)\frac{(3x)^2}{2!} + (-1)(-2)(-3)\frac{(3x)^3}{3!}$ $1 - 3x + 9x^2 - 27x^3$	M1(AO 1.1) A1(AO 1.1) A1(AO 1.1) [3]	allow sign errors allow recovery from omission of brackets		
		b	$ x < \frac{1}{3}$	B1(AO 1.1) [1]	$-\frac{1}{3} < x < \frac{1}{3}$		
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
50		a	$\frac{16}{3}$ [units]	B1 (AO 2.2a) [1]	oe		
		b	At each iteration, the perimeter is $\frac{4}{3}$ times the previous perimeter Perimeter of n th = $3 \times \left(\frac{4}{3}\right)^n$ iteration	M1 (AO 3.1a) A2 (AO 1.1, 2.2a) [1]	May be a specific instance Allow A1 for 3 times a power of $\frac{4}{3}$		
		c	As $n \rightarrow \infty$, $3 \times \left(\frac{4}{3}\right)^n \rightarrow \infty$	E1 (AO 2.4) [1]	o $3 \times \left(\frac{4}{3}\right)^n$ r increases without limit as n increases		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
51		a	Area of each little = $\frac{1}{9}$ unit triangle Area = $\frac{1}{3}$ [units] a	M1 (AO 2.2a) A1 (AO 1.1) [1]	or total additional are = $\frac{1}{3}$ a		
		b	$\frac{12}{81}$	B1 (AO 2.2a) [1]	oe	$\frac{4}{27}$	
		c	$1 + \frac{3}{9} + \frac{4 \times 3}{9 \times 9} + \frac{4 \times 4 \times 3}{9 \times 9 \times 9} + K$ $1 + \frac{\frac{3}{9}}{1 - \frac{4}{9}}$ $\frac{8}{5}$ [units]	M1 (AO 3.1a) M1 (AO 2.2a) M1 (AO 3.1a) A1 (AO 1.1) [5]	4 times for new number of triangles Triangle area divided by 9 Use of GP formula		
			Total	7			

Question		Answer/Indicative content	Marks	Part marks and guidance		
52	a	$1 + (-3)(-ax) + \frac{(-3)(-4)}{2}(-ax)^2 + \dots$ Equate coefficients $3a = 6a^2$ $a = \frac{1}{2}$	M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO1.1b) [3]	Attempt to use the binomial expansion Equating their coefficients oe www	Allow sign errors, bracket errors, a slip Allow recovery from missing brackets Their equation should not involve x	
Examiner's Comments A number of candidates could not apply the binomial expansion with a negative power, however the majority were able to expand sufficiently well to earn the method mark. Algebraic slips, lack of care with negative signs and failing to use brackets led to most of the mistakes seen. The second method mark was given for equating coefficients, so many candidates lost 2 marks here even where the value $a = 2$ was seen.  Misconception The word coefficient was seemingly not understood by many candidates who equated terms rather than coefficients. Candidates who subsequently lost one or more x from their equation were not given full marks here.						

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b	i	Valid for $ x < 2$	B1 (AO 2.3) [2]	Acc $x < \frac{1}{ a }$ for ept r their a Do not accept $x < 2$, $\left \frac{1}{2}x\right < 1$ or x ≤ 2 or similar Examiner's Comments About half the candidates realised that $x < 2$ was required here, but many lost the mark for omitting the modulus or using $\leq$. $x < \frac{1}{2}$ was also often seen.	
		ii	$\left[\left(1 - \frac{1}{2}x\right)^{-3}\right] \approx 1 + \frac{3}{2}x + \frac{3}{2}x^2$	B1 (AO 1.1b) [1]	Cao Examiner's Comments Even when the value of a was correct, many candidates did not realise what was required here. Those candidates who had values other than 2 for a did not seem to realise their error when the coefficients of x and x^2 in their expansion were not equal.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
53	a		3	B1 (AO3.3) [1]	<u>Examiner's Comments</u> Candidates who did well in this question recognised the geometric progression and completed the parts successfully.	
	b		$[8 \times 3^4 =] 648$	B1 (AO1.1) [1]	<u>Examiner's Comments</u> This part was also well answered by the majority of candidates.	
	c		$\frac{8(3^n - 1)}{3 - 1}$ $= 4(3^n - 1) \text{ or } -4(1 - 3^n)$	B1 (AO3.4) [1]	use of formula for sum of gp mark the final answer or $4 \times 3^n - 4$ <u>Examiner's Comments</u> Candidates who did less well on this part either didn't attempt to simplify their answer or they simplified it incorrectly, 12^n was frequently seen.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		their $4(3^n - 1) = 185\,207$ or $3^n = 46303(.75)$ awrt 9.8 cao [=] 9	M1 (AO3.1a) A1 (AO1.1) A1 (AO3.2a) [3]	M0 for eg $8 \times 3^{n-1}$ no FT available here not from wrong working allow use of $<$ or $\leq$ for up to 3 marks allow M1 only for use of $>$ or $\geq$ or $3^9 = 19683$ and $3^{10} = 59049$ seen for M1 then A1 (compariso n with 46 303)	
					<u>Examiner's Comments</u> Candidates who did well in this part recognised that the request to 'Determine the maximum number of days the model could be viable' necessitated showing supporting calculations and formed a correct equation or inequality. Some candidates were unable to solve their equation, or made slips with manipulating the inequality.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	e		unlikely because eg some of the population will be immune to the virus eg some of the population will not be exposed to the virus eg medical intervention eg extrapolation eg movement of people in and out of town	B1 (AO3.5b) [1]	any sensible reason it's unlikely that everyone will be affected or is insufficient Examiner's Comments Those candidates that were successful with the above part were usually successful in explaining their result in context. A satisfactory reason for it being unlikely that the model would be viable for this long usually ensued. Sometimes a circular argument was presented in this part		
			Total	7			

Question	Answer/Indicative content	Marks	Part marks and guidance		
54	Let $u = 1 + \sqrt{x}$ $\Rightarrow dx = 2(u - \frac{1}{2}x^{-\frac{1}{2}}) du$ or $1) du$ or $2\sqrt{x} du$ $\int \frac{x}{1+\sqrt{x}} dx = \int \frac{2(u-1)^3}{u} du$ $\int \frac{2(u^3 - 3u^2 + 3u - 1)}{u} du$ $\int 2(u^2 - 3u + 3 - \frac{1}{u}) du$ $\frac{2u^3}{3} - 3u^2 + 6u - 2\ln u + c$ $\frac{2(1+\sqrt{x})^3}{3} - 3(1+\sqrt{x})^2 + 6(1+\sqrt{x}) - 2\ln 1+\sqrt{x} + c$ oe	M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO2.5) [6]	In terms of u For expanding cubic (allow one error) Dividing by u 3 of <i>their</i> terms correctly integrated to inc $\ln u$. Constant term may be missing at this stage Correct answer in terms of x with constant term Examiner's Comments The more able candidates often produced fully correct work but some lost the factor of 2 after integrating. The expansion of $(u-1)^3$ was often inaccurately done.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Some candidates wasted time expanding brackets and attempting to factorise at the end – no particular form was requested so candidates could leave their answer unsimplified. Some candidates attempted to use integration by parts rather than simplifying the integral by expanding brackets and dividing by u.	
			Total	6		

Question		Answer/Indicative content	Marks	Part marks and guidance		
55	a	Arithmetic sequence with $a=9300$ $a_{10} = a + 9d = 9300 + 9d = 3900$ $d = -600$ $a_{20} = a + 19d = 9300 - 19 \times 600 = -2100$ [So 20th term is negative]	M1 (AO 3.01a) A1 (AO 1.1b) A1 (AO 2.1) [3]	Using formula for term of AP with values substituted soi Allow for -2100 without comment	eg $d = 600$ stated but later subtracted Also allow for earlier negative term found and comment that 20th is less	
	b	S is increasing as long as the extra terms are positive First negative term when $a_n < 0$ $9300 - 600(n - 1) < 0 \Rightarrow$ $\frac{9300}{600} < n - 1 \Rightarrow n > 16.5$ So maximum sum after 16 terms	M1 (AO 3.1a) A1 (AO 2.2a) M1 (AO 1.1a) A1 (AO 1.1b) [4]	Attempt to find first negative term with 9300 and their d Allow $n > 16$ or $n \geq 17$ oe Using sum with their n and their d cao	(or last positive term) Also allow M1A1 for establishing $a_{16} = 300$ and $a_{17} = -300$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			$S_{16} = \frac{16}{2}(2 \times 9300 - 600(16 - 1))$ $= 76\,800$ Alternative solution $S = \frac{n}{2}(2 \times 9300 - 600(n - 1)) \quad [= 9600n - 300n^2]$ [To find max S treating S and n as continuous] $\frac{dS}{dn} = 9600 - 600n = 0$ Max sum when $n = 16$ $S_{16} = \frac{16}{2}(2 \times 9300 - 600(16 - 1)) = 76\,800$ Second alternative solution $S_{15} = 76\,500$ $S_{16} = 76\,800$ $S_{17} = 76\,500$ max total is 76800	M1 M1 A1 A1 M1 M1 A1 A1	Using the sum formula with 9300 and their d substituted Setting derivative to zero and solve Allow for $n = 16$ Cao Using the sum formula with 9300 and their d to find at least two totals Evaluating S_{15}, S_{16} and S_{17} Award for S_{16} as maximum identified from correct working Cao	Also allow M1 for other methods for finding the max A1 for $n = 16$ eg M1A1 for $S_n = k - 300(n - 16)^2$ Allow BC FT their d: three consecutive totals around their maximum eg $d = -540$ needs 17th, 18th, 19th totals	Examiner's Comments Whilst part (a) was designed as a hint for this second part of the question, a significant number of candidates ignored the hint and proceeded with the first

Question			Answer/Indicative content	Marks	Part marks and guidance
					alternative method successfully. Some who were attempting to find the first negative term often tried to solve $\text{total} = 0$ rather than $\text{term} = 0$. Some good answers were seen from candidates who had used lists of values from their calculator.  Afl It is possible to create lists of values for terms or totals on a calculator that can be the basis of a good solution. Many candidates however did not show sufficient evidence to make a sound conclusion. The minimum required here was clearly labelled values for the total immediately either side of the maximum value seen; ideally an argument that the formula for total is quadratic so the maximum found is the maximum globally, or that the total will keep decreasing as the terms will remain negative should be given.
			Total	7	

Question			Answer/Indicative content	Marks	Part marks and guidance		
56			$\frac{f(x+h)-f(x)}{h} = \frac{(x+h)^3 - (x^3 - x)}{h}$ $= \frac{x^3 + 3hx^2 + 3h^2x + h^3 - x^3 + x}{h}$ $= \frac{3x^2h + 3xh^2 + h^3 + x}{h} = 3x^2 + 3xh + h^2 + 1$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ $= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 + 1) = 3x^2 + 1$	M1(AO2.1) A1(2.1) M1(AO2.1) E1(AO2.1) [4]	Substituting $\frac{f(x+h)-f(x)}{h}$ and into attempt to expand $(x+h)^3$ Correct expansion of $(x+h)^3$ Simplifying the fraction to eliminate a denominator Must include the idea of limit as h tends to zero AG	Allow correct 6 terms not simplified	
			Total	4			
57			BC $2 \times 0 + 4 \times 1 + 8 \times 2 + 16 \times 3 + 32 \times 4$ 196	M1 (AO1.1a) A1 (AO1.1) [2]	At least two terms written out, allow one error, OR correct answer	Terms may not be seen as may be done completely on calculator	
			Total	2			

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
58			$10 + 10 \times 0.98 + 10 \times 0.98^2$ or $10 + 9.8 + 9.604$ $\frac{10}{1-0.98}$ 500	M1 (AO3.1b) M1 (AO1.1) A1 (AO3.2a) [3]	Use of GP with common ratio 0.98 Sum to infinity	GP implied by formula for S_{∞} , S_n or un showing 0.98 or 1 st 3 terms	
			Total	3			
59	a		26	B1 (AO1.1) [1]	NB $17 + (n - 1) \times 11 = 292$		
	b		$\frac{\text{their } 26}{2} \times (17 + 292)$ oe 4017	M1 (AO1.1) A1 (AO1.1) [2]	$\frac{\text{their } 26}{2} \times (2 \times 17 + (\text{their } 26 - 1) \times 11)$ Examiner's Comments Question 5 was generally very well done. Candidates who did less well made a slip in part (a) – usually giving n as 25 – and then went on to earn the method mark in part (b).		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
60	a		$1 + \left(\frac{1}{2}\right)(4x) + \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\frac{(4x)^2}{2!}$ $1 + 2x - 2x^2$ isw cao	M1 (AO2.1) A1 (AO1.1) A1 (AO1.1) [3]	ignore extra terms, allow one error two of three terms correct all three terms correct, ignore extra terms	if M0 allow SC2 for 2 of first three terms correct	
	b		$ x < \frac{1}{4}$ oe	B1 (AO1.1) [1]	or $ x \leq \frac{1}{4}$ oe Examiner's Comments Question 6 was generally very well done. Candidates who did less well made a slip in the expansion – usually in the third term – or were unable to answer part (b) correctly.		
			Total	4			